

NEXRAD EXPANDED SITE SURVEY AND ENVIRONMENTAL ASSESSMENT REPORT FOR THE OPERATIONAL SUPPORT FACILITY-3 (OSF-3) NORMAN, OKLAHOMA

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ABSTRACT

This report documents the findings of the NEXRAD site survey for the Operational Support Facility-3 (OSF-3) site, Norman, Oklahoma. The OSF-3 site is recommended for selection.

The significant recommendations and conclusions of this report are as follows:

Site Location

Radar Site Max Westheimer Airpark, Norman, Oklahoma.

Site Configuration The radar and the OSF office (NEXRAD principal user facility) would be connected by a coaxial cable.

NEXRAD Tower Geographic Coordinates Latitude = 35° 14' 18" N; longitude = 97° 27' 36" W (*North American Datum, 1927*).^{*}

Site Description A 1-acre parcel in NE1/4 SE1/4, Section 24, T. 9 N., R. 3 W., Indian Baseline and Meridian.

Recommended NEXRAD Tower Height (without radome) 30 m AGL.

Recommended NEXRAD Frequency Assignment 2950 MHz.

Environmental Impact A site-specific Environmental Impact Statement (EIS) is not considered necessary at this time.

Probability of FAA Approval of NEXRAD Tower Good.

Other Significant Findings

The proposed site for construction of NEXRAD will be leased by the NWS.

The City of Norman has been informed of the planned construction of NEXRAD at the proposed site.

The lead time for providing electrical power service to the proposed NEXRAD site may be as long as six months.

Waveguide bandpass filters would be required in both the KOUN and OSF-3 NEXRAD systems.

Filtering or the addition of a metal screen may be required to suppress interference from the OSF-3 NEXRAD in the upper MLOS antenna that supports the data link between OSF and the Oklahoma City NEXRAD at Twin Lakes.

^{*}Equivalent to Latitude: 35° 14' 18" N; Longitude: 97° 27' 37" W (*North American Datum, 1983*).

The environmental assessment contained within this report concludes that no significant environmental impact would result from construction and operation of the proposed facilities. A Finding of No Significant Impact (FONSI) is recommended.

In the unlikely event that archaeological materials are uncovered during construction, the Oklahoma Archaeological Survey office should be notified immediately.

The final *Programmatic Environmental Impact Statement* (PEIS) and the final *Supplemental Environmental Assessment* (SEA) for the NEXRAD program have been published. To obtain copies of the documents, please see the preface of this report.

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ABBREVIATIONS

AAIA	Airport and Airway Improvement Act
AES	Area Electronic Supervisor
AFB	Air Force Base
A/G	air/ground
AGL	above ground level
AM	Area Manager
AP	anomalous propagation
ARSR	Air Route Surveillance Radar
ARTCC	Air Route Traffic Control Center
ASDE	Airport Surface Detection Equipment
ASL	above sea level
ASR	Airport Surveillance Radar
ATC	Air Traffic Control
ATCBI	Air Traffic Control beacon interrogator
ATCT	Air Traffic Control tower
AWS	Air Weather Service
BLM	Bureau of Land Management
DF	direction finder
DME	distance measurement equipment
DoC	Department of Commerce
DoD	Department of Defense
DoT	Department of Transportation
DVIP	Digital Video Integrated Processing
ECAC	Electromagnetic Compatibility Analysis Center
EIS	Environmental Impact Statement
EMI	electromagnetic interference
FAA	Federal Aviation Administration
FAR	Federal Aviation Regulations
FAC	Final Approach Course
FONSI	Finding of No Significant Impact
HVAC	heating, ventilating, air conditioning
IFR	instrument flight rules
ILS	instrument landing system
INR	interference-to-noise ratio
ISA	initial site assessment
ISS	In-depth Site Survey
JSPO	Joint System Program Office
LOS	line of sight
MDA	minimum descent altitude

ABBREVIATIONS (continued)

MDS	minimum discernible signal
MIC	Meteorologist-in-Charge
MLOS	microwave line of sight
MLS	microwave landing system
MSL	mean sea level
NAS	Naval Air Station
NDB	nondirectional beacon
NEC	National Electric Code
NEPA	National Environmental Policy Act
NEXRAD	Next Generation Weather Radar
NOAA	National Oceanic and Atmospheric Administration
NOC	Naval Oceanography Command
NTR	NEXRAD Technical Requirements
NWS	National Weather Service
OIC	Official-in-Charge
OSF	Operational Support Facility
PEIS	Programmatic Environmental Impact Statement
PPI	plan position indicator
PRF	pulse-repetition frequency
PSS	Preliminary Site Survey
PUF	principal user facility
PUP	principal user processor
RACON	radar transponder beacon
RCAG	Remote Communications Air-Ground Facility
RDA	radar data acquisition
RF	radio frequency
RFC	River Forecast Center
RFI	radio frequency interference
RPG	radar product generation
RPGOP	radar product generation operational PUP
RTR	remote transmit/receive site
SEA	Supplemental Environmental Assessment
TACAN	tactical air navigation
TERPS	U.S. Standard Terminal Instrument Procedures
UBC	Uniform Building Code
USFWS	U.S. Fish and Wildlife Service
USFS	U.S. Forest Service
USGS	U.S. Geological Survey
VHF	very high frequency

ABBREVIATIONS (concluded)

VLA	very large array
VLBA	very long baseline array
VOR	very high frequency omnirange
VORTAC	very high frequency omnirange tactical air navigation
WFO	Weather Forecast Office

PREFACE

A new nationwide network of weather radars called the Next Generation Weather Radar (NEXRAD) is planned to replace the existing system of weather radars. The purpose of the NEXRAD system is to reduce weather-related loss of life, injury, property damage, and interruption of economic activity by the timely acquisition, processing, and distribution of weather radar information. The principal users of the NEXRAD weather information will be the National Weather Service (NWS), the Federal Aviation Administration (FAA), the Air Weather Service (AWS), and the Naval Oceanography Command (NOC). The first production NEXRADs were scheduled to be installed during 1989; installation will continue through the mid-1990s.

SRI conducted an evaluation to design a coherent, integrated network of nationwide radar coverage tailored to the unique requirements of the user agencies. Radar coverage and the fulfillment of the users' requirements were the governing site-selection factors. Existing weather radar sites were given first consideration for selection. To avoid long communications links, radars were collocated with existing NWS offices when practical. When new locations were selected, sites on government-owned land were given priority consideration over sites on privately owned land. Site preparation costs were also an important factor.

This report documents the survey findings for Operational Support Facility-3 (OSF-3). During this site survey, the site was evaluated both for radar location and construction feasibility. During another survey visit, an instrumentation van was used to take up to 48 continuous hours of radio-frequency (RF) measurements to aid in selection of a frequency assignment. In addition, a specially designed video system in the van aided in determining an optimum height for the radar antenna; and an X-band radar in the van was used to determine the ground clutter extent and intensity.

The final *Programmatic Environmental Impact Statement* (PEIS) for the NEXRAD program was published in November 1984. The final *Supplemental Environmental Assessment* (SEA) that updates information in the final PEIS was published in April 1993. Copies of the final PEIS and the final SEA may be obtained by written request to:

Ms. Elizabeth F. Ewell
NEXRAD JSPO
National Weather Service
1325 East West Highway, SSMC-2
Room 15440
Silver Spring, MD 20910.

Supporting analyses of the recommended frequency assignments by the Electromagnetic Compatibility Analysis Center (ECAC) are contained in Appendix A.

Pertinent correspondence and background information are contained in Appendix B. As a convenience, Appendix C contains names, addresses, and telephone numbers of contact points in local organizations and companies. Appendix D discusses project alternatives considered and why they were rejected.

I INTRODUCTION

A. General Description

The proposed site to serve the NEXRAD Operational Support Facility (OSF) is at the Max Westheimer Airpark, Norman, in Cleveland County, Oklahoma. The OSF office (approximately 300 ft south of the proposed radar site) would be the NEXRAD principal user facility (PUF) where the NEXRAD display and control equipment would be operated. A coaxial cable would connect the radar to the PUF.

The proposed NEXRAD site was selected for the following reasons:

- The site is adjacent to OSF.
- A radar at this site would provide excellent coverage in all directions.
- The site is adjacent to an existing road.
- The site is adjacent to existing power lines.

Location plans, a conceptual site layout, and photographs are shown in Figures 1 through 3. Figure 4 presents a panoramic view of the horizon taken by the NEXRAD van video system at the proposed radar site. The video camera was at 114 ft AGL, which is equivalent to the recommended height to the center of the NEXRAD antenna.

B. Ownership and Availability

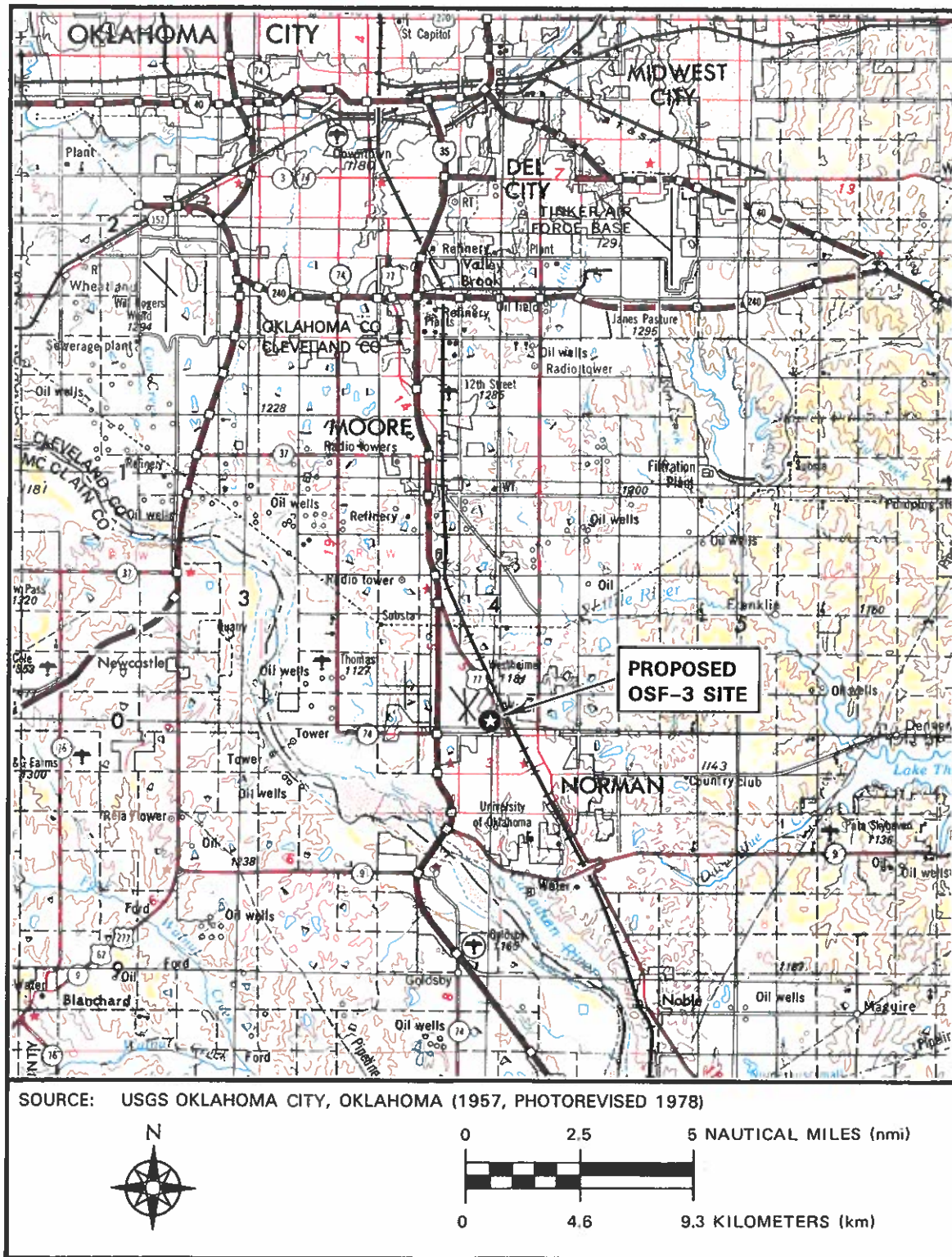
The proposed radar site is owned by the University of Oklahoma. The proposed site is available for lease (George Hargett, personal communication, August 1994).

C. Site Size and Acquisition

The proposed radar site is approximately 1 acre because that is the optimum lot size for constructing a NEXRAD on a level site.

D. Present Use

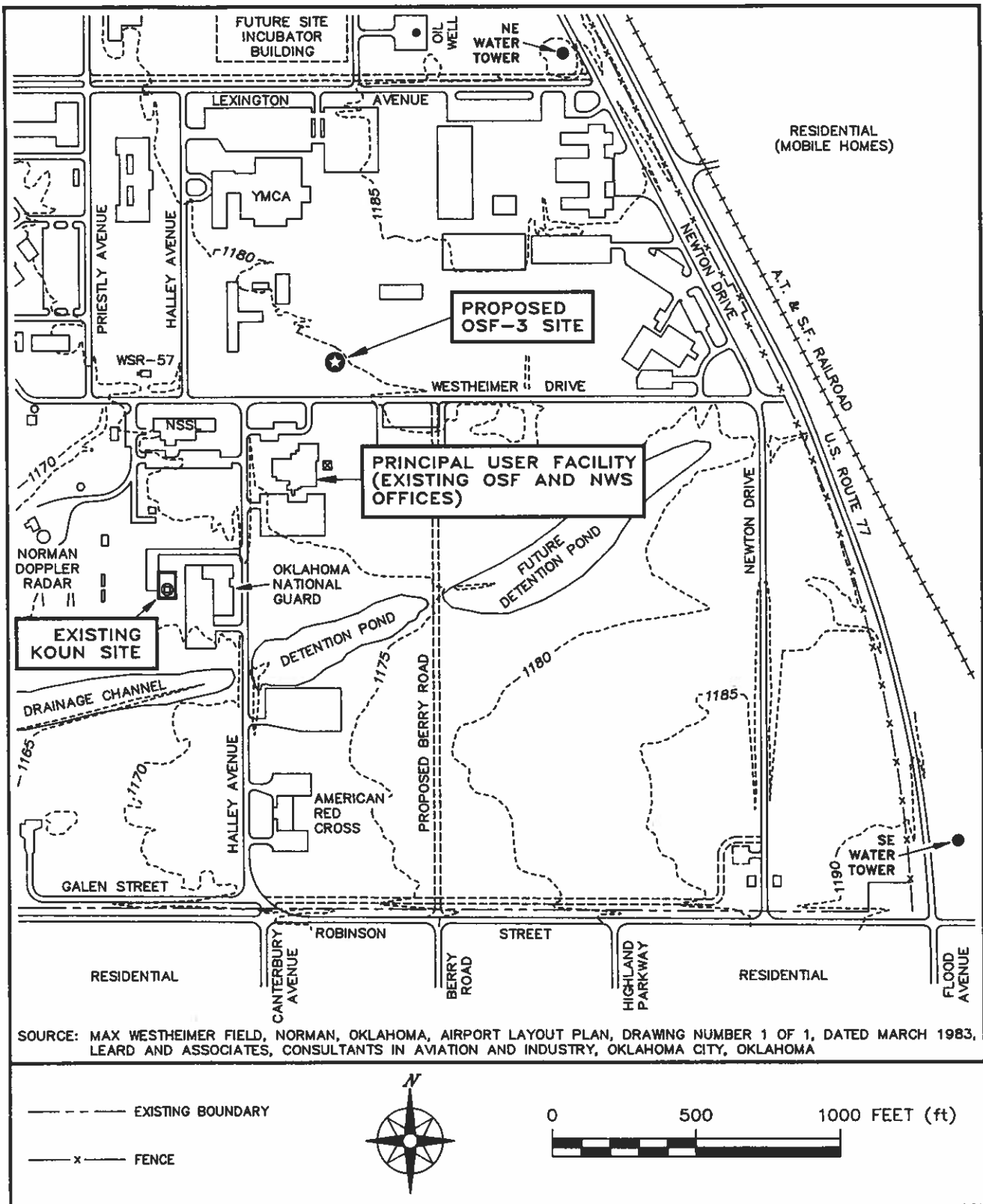
The proposed site is an unused, open grass-covered field. No buildings are on the site. [See Figures 3(a) and 3(b)].



(a) PROPOSED FACILITIES — 1:250,000 SCALE

FIGURE 1 SITE LOCATION PLAN — PROPOSED OSF-3 SITE AT MAX WESTHEIMER AIRPARK, NORMAN, OKLAHOMA

FIGURE 1 SITE LOCATION PLAN — PROPOSED OSF-3 SITE AT MAX WESTHEIMER AIRPARK, NORMAN, OKLAHOMA



(c) PROPOSED FACILITIES - 1:6000 SCALE

FIGURE 1 SITE LOCATION PLAN - PROPOSED OSF-3 SITE AT MAX WESTHEIMER AIRPARK, NORMAN, OKLAHOMA

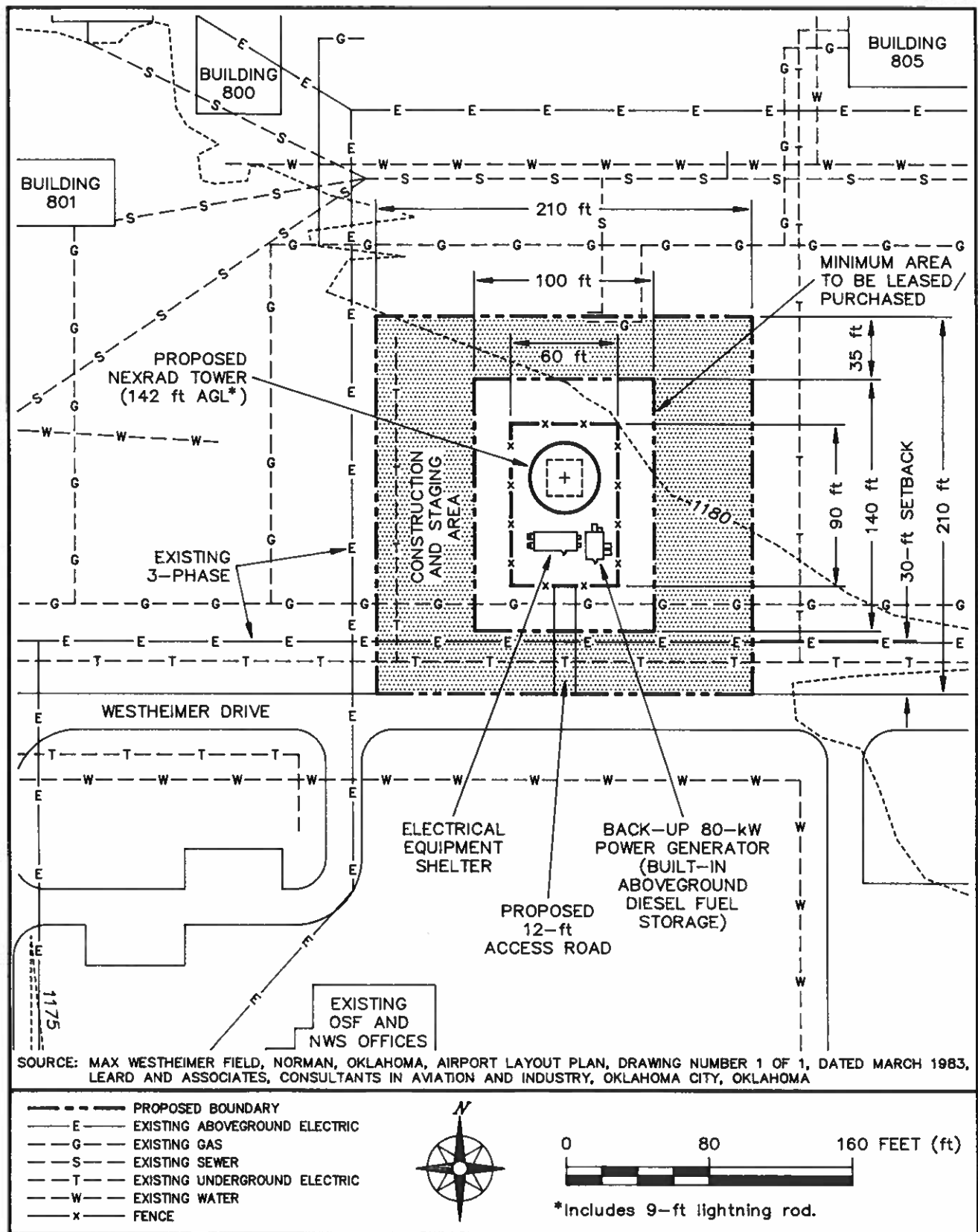


FIGURE 2 CONCEPTUAL SITE LAYOUT AND CONFIGURATION PLAN — PROPOSED OSF-3 SITE AT MAX WESTHEIMER AIRPARK, NORMAN, OKLAHOMA



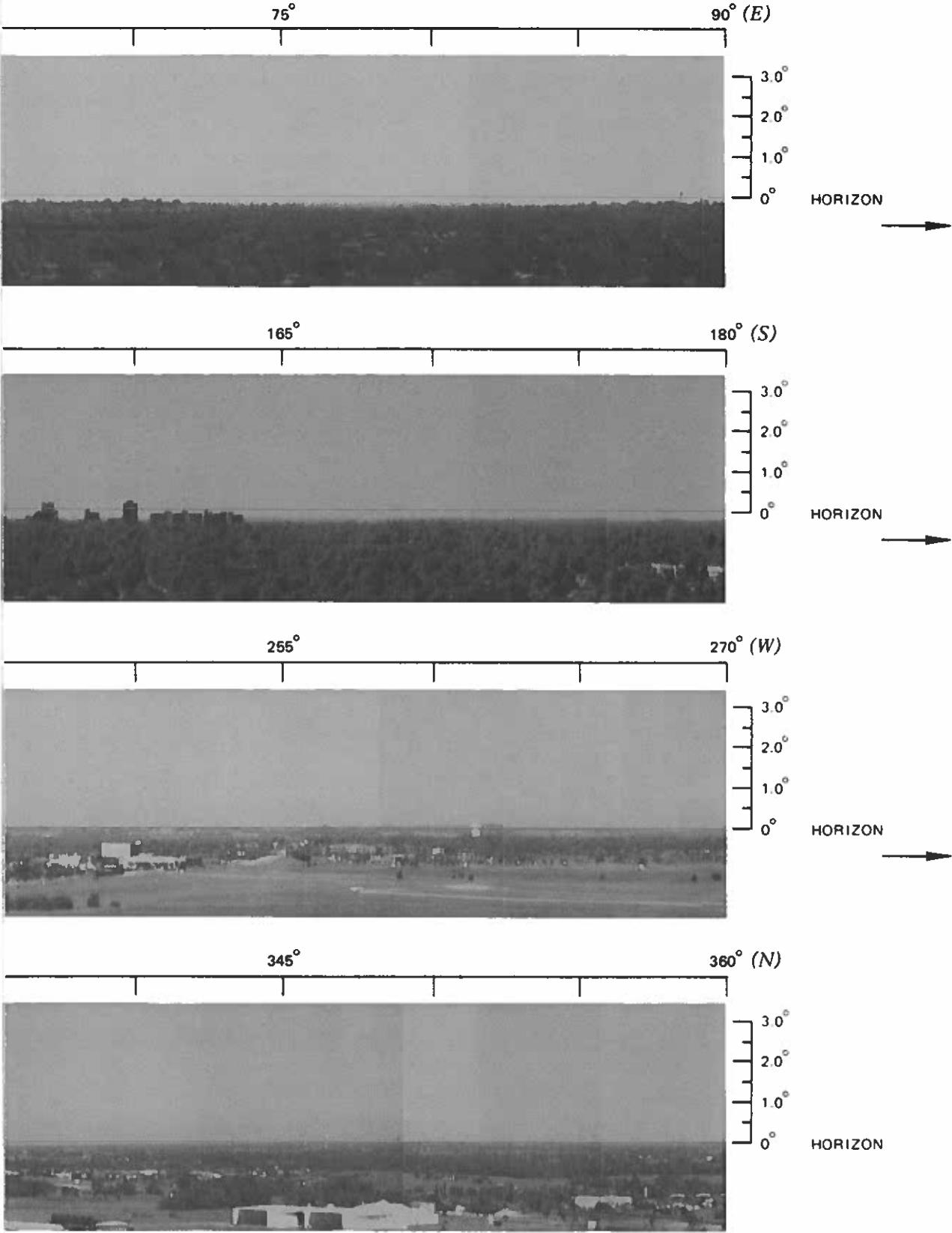
(a) VIEW (LOOKING EAST) OF PROPOSED OSF-3 SITE

FIGURE 3 SITE PHOTOGRAPHS -- PROPOSED OSF-3 FACILITIES AT MAX WESTHEIMER AIRPARK, NORMAN, OKLAHOMA



(b) VIEW (LOOKING SOUTHWEST) OF PROPOSED OSF-3 SITE

FIGURE 3 SITE PHOTOGRAPHS — PROPOSED OSF-3 FACILITIES AT MAX WESTHEIMER AIRPARK, NORMAN, OKLAHOMA (concluded)



OKLAHOMA

FIGURE 4

E. Climate

See the *1988 Local Climatological Data, Annual Summary with Comparative Data, Oklahoma City, Oklahoma*, in Appendix B.

F. Topography and Geographic Coordinates

The proposed radar site is nearly flat. The geographic coordinates of the NEXRAD tower would be: latitude = 35° 14' 18" N; longitude = 97° 27' 36" W (*North American Datum, 1927*);* ground elevation = 1,180 ft MSL.

The site description is as follows: a 1-acre parcel in NE1/4 SE1/4, Section 24, T. 9 N., R. 3 W., Indian Baseline and Meridian.

G. Soils

Soils at this site have been identified as Doolin-Urban Land-Pawhuska complex. Characteristics of this soil type show it as a grayish brown silt loam about 9 inches thick with the subsurface layer a light brownish gray silt loam to a depth of about 12 inches. Doolin soil is high in natural fertility, where Pawhuska soil is low in natural fertility and organic matter. This soil has a high shrink-swell potential, very slow permeability, high sodium and salt content, high corrosion hazard, high erosion hazard, and low strength.†

See Appendix B for additional soils information.

H. Floodplain

The proposed site is not within a 100-year floodplain [FEMA Flood Insurance Rate Map (FIRM), City of Norman, Oklahoma, Cleveland County; Community Panel Number 400046 0015 B (November 1, 1979)].

I. Seismic Zone

The Norman, Oklahoma, area is classified Zone 2, moderate damage. Zones range from 0 to 4, with 0 indicating that the area has no potential for damage, and 4 indicating the area has a potential for great damage.‡

*Equivalent to Latitude: 35° 14' 18" N; Longitude: 97° 27' 37" W (*North American Datum, 1983*).

†U.S. Department of Agriculture, Soil Conservation Service in Cooperation with Oklahoma Agricultural Experiment Station, *Soil Survey of Cleveland County, Oklahoma* (1983).

‡"General Design Requirements, Earthquake Regulations, Seismic Zones," International Conference of Building Officials, *Uniform Building Code* (May 1991).

J. Surrounding Terrain

Immediately adjacent to the radar is the University of Oklahoma Max Westheimer Airpark and University of Oklahoma Research Park. The Max Westheimer runways lie about 3,000 ft northwest of the proposed site. Downtown Norman and the University of Oklahoma are about 2 nmi southeast of the site. Will Rogers World Airport and Tinker Air Force Base are about 12 nmi to the northwest and northeast, respectively. Oklahoma City is about 16 nmi to the north. Communities and farmland extend in all directions.

K. Future Development Plans

Based on currently available information, no major development is anticipated near the proposed radar site (Douglas J. Koscinski, personal communication, May 1994).

II RADAR COVERAGE

A. Weather Approach Directions

Weather fronts generally approach the Norman area from the west. Figure 5 shows the weather approach directions of interest. Generally, unobstructed radar coverage toward these directions is more important than toward other directions. Seasonal variations are summarized below.

Winter Winter fronts generally approach from the west; however, severe ice and heavy snowstorms approach from the northwest.

Spring In spring, fronts move from west-northwest; the precipitation cells embedded in the frontal system generally move from the southwest at a maximum speed of 60 knots. Spring usually brings floods, which are the result of heavy rain. In Oklahoma City and other urban communities, flooding can be caused by rapid runoff of heavy rains. Tornadoes are most frequent during this season.

Summer Summer fronts (mainly Pacific fronts) and cells from air-mass thunderstorms approach from the west and southwest at a maximum of 70 knots.

Fall In the fall, the approach shifts slightly to the northwest, with embedded storm cells moving from the southwest. Flash floods can occur during this season when decaying tropical storms enhance the moisture content in storm cells.

The number of occurrences of severe weather per year within the Oklahoma City area is listed in Table 1. As can be seen from the list, severe weather can occur any time during the year (Larry Mooney, personal communication, May 1985).

B. Radar Coverage

Radar coverage diagrams for 2,000, 4,000, and 6,000 ft above the radar site elevation are shown in Figure 6. The radar coverage diagrams are overlaid on a map showing surrounding populated areas, airports identified by the FAA as important, and priority Department of Defense (DoD) facilities. The coverage plots were computer-generated by ECAC. The ECAC data base uses a 30-sec terrain grid that does not consider the effects of trees and buildings or other man-made obstructions. These coverage plots somewhat overstate the radar coverage in instances where obstructions are not contained in the ECAC data base. When necessary, the plots are manually updated by SRI International using additional information gathered at the site.

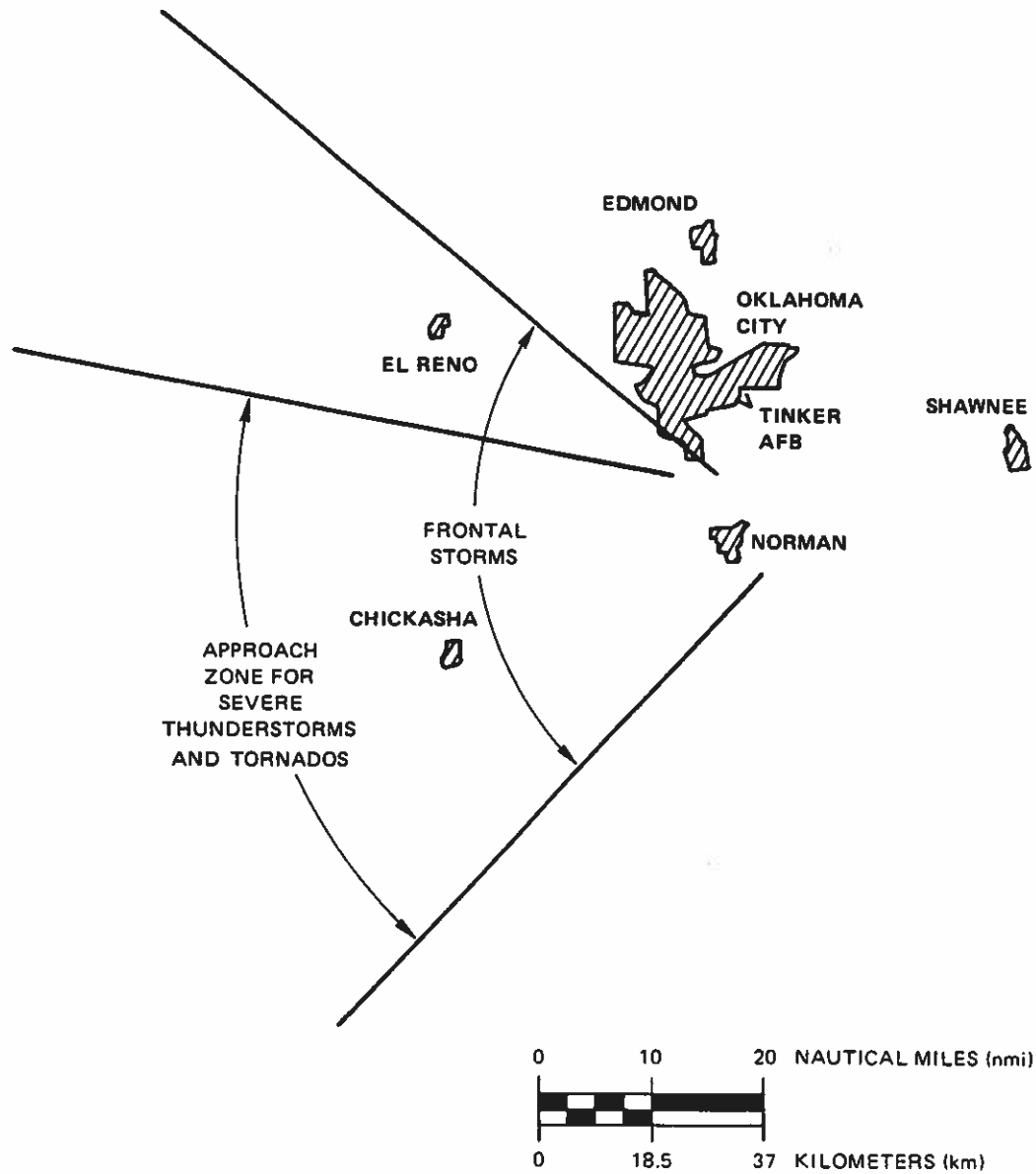


FIGURE 5 SEVERE WEATHER APPROACH ZONES — NORMAN, OKLAHOMA, AREA

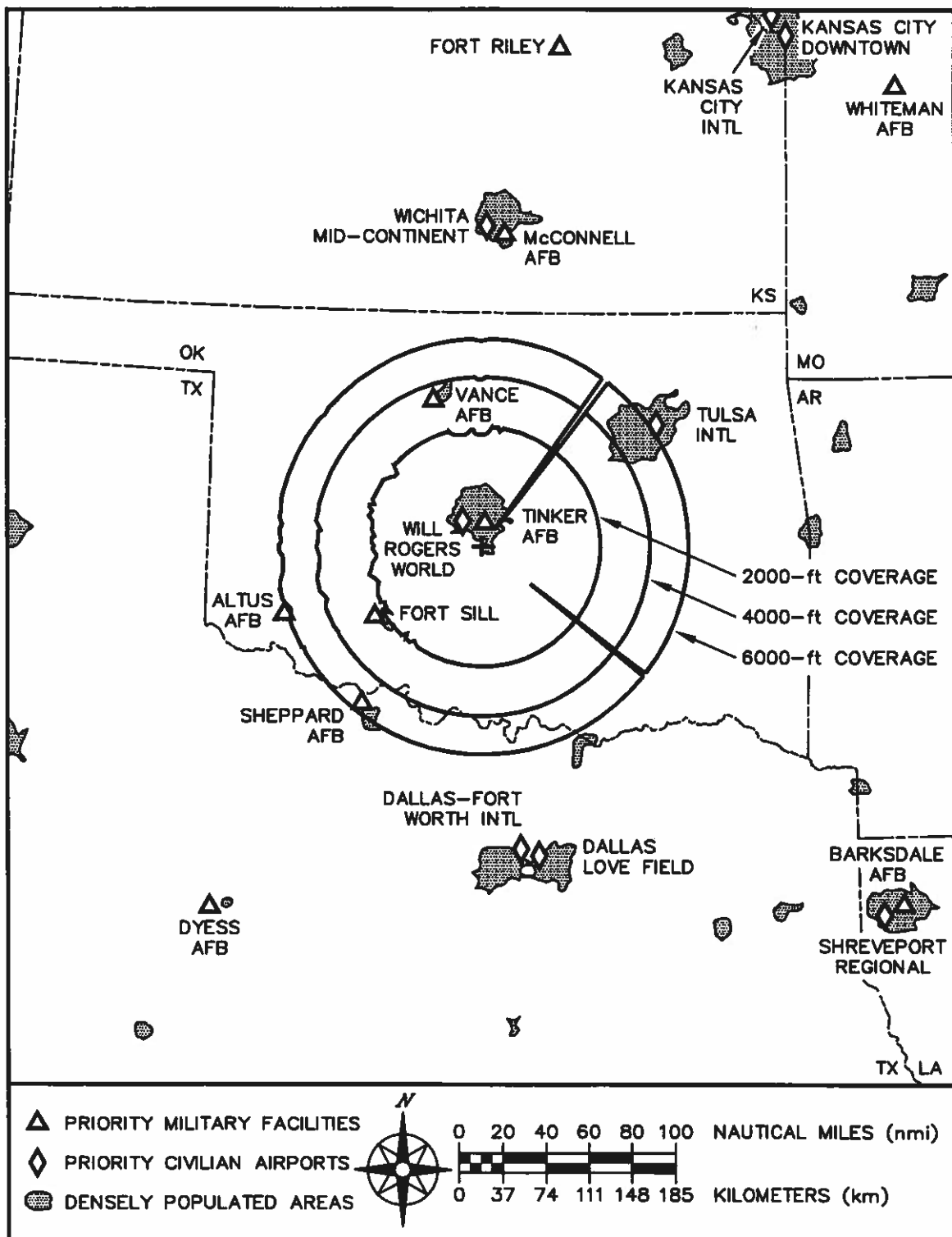


FIGURE 6 ESTIMATED RADAR COVERAGE - PROPOSED OSF-3 SITE
 AT MAX WESTHEIMER AIRPARK, NORMAN, OKLAHOMA

Table 1
OCCURRENCES OF SEVERE WEATHER

Category	Average Occurrences Per Year	Time of Year
Tornados	50	Spring, Fall
Severe Thunderstorms	50	Spring, Summer, Fall
Flash Flood Storms	10	Spring, Fall
Wind Downburst/Microburst	50	Spring
Ice Storms	3	Winter, Early Spring
Blizzards	9	Winter

As can be seen, coverage in all directions would be unobstructed except for a couple of narrow notches. The notches in coverage toward the northeast and southeast are caused by nearby water towers.

C. Clutter Estimate

The extent to which clutter would degrade radar coverage at this proposed site has been estimated for three cases:

Case 1 This case considers all clutter returns with intensity greater than 0-dBZ weather returns (corresponding to rain falling at 0.0015 inch/hr).

Case 2 This case considers only clutter returns with intensity greater than +29-dBZ weather returns (corresponding to rain falling at 0.1 inch/hr, which is equivalent to digital video integrated processing [DVIP] level-2 of the WSR-57, WSR-74S, and WSR-74C radars).

Case 3 This case considers only clutter returns with intensity greater than +44-dBZ weather returns (corresponding to rain falling at 0.9 inch/hr, which is nearly equivalent to DVIP level-4 of the WSR-57, WSR-74S, and WSR-74C radars).

These estimates are based on measurements made by the SRI NEXRAD van using an X-band marine radar at the OSF-3 site. The measurements were scaled to reflect the specifications of the *NEXRAD Technical Requirements* (NTR).^{*} The estimates for Cases 1, 2, and 3 are presented in Figures 7, 8, and 9. The figures show major population

^{*} The clutter cancellation specified in the NTR can be selected—20, 30, or 50 dB. We use 30 dB as an intermediate value.

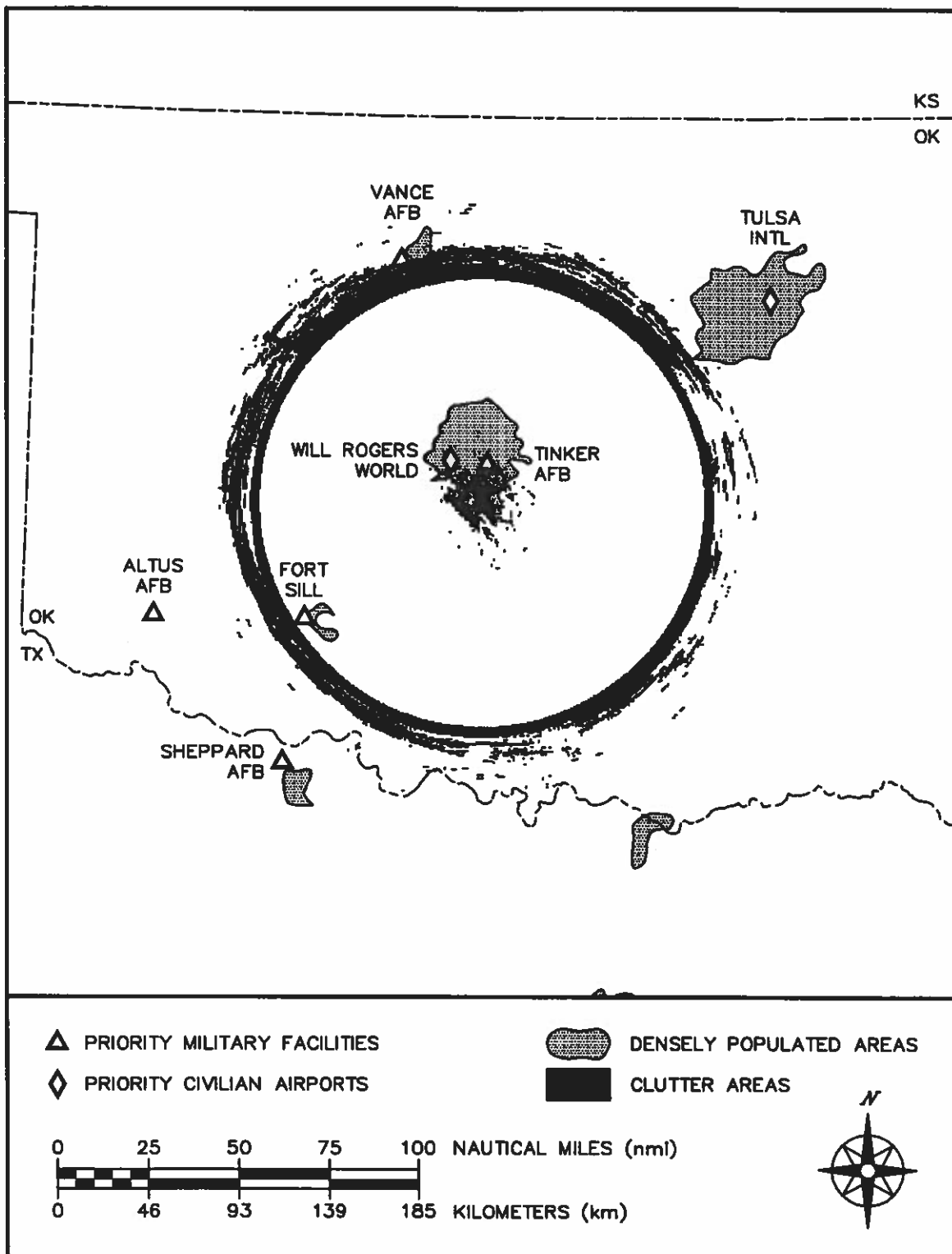


FIGURE 7 CLUTTER ESTIMATE, > 0 dBZ - PROPOSED OSF-3 SITE
 AT MAX WESTHEIMER AIRPARK, NORMAN, OKLAHOMA

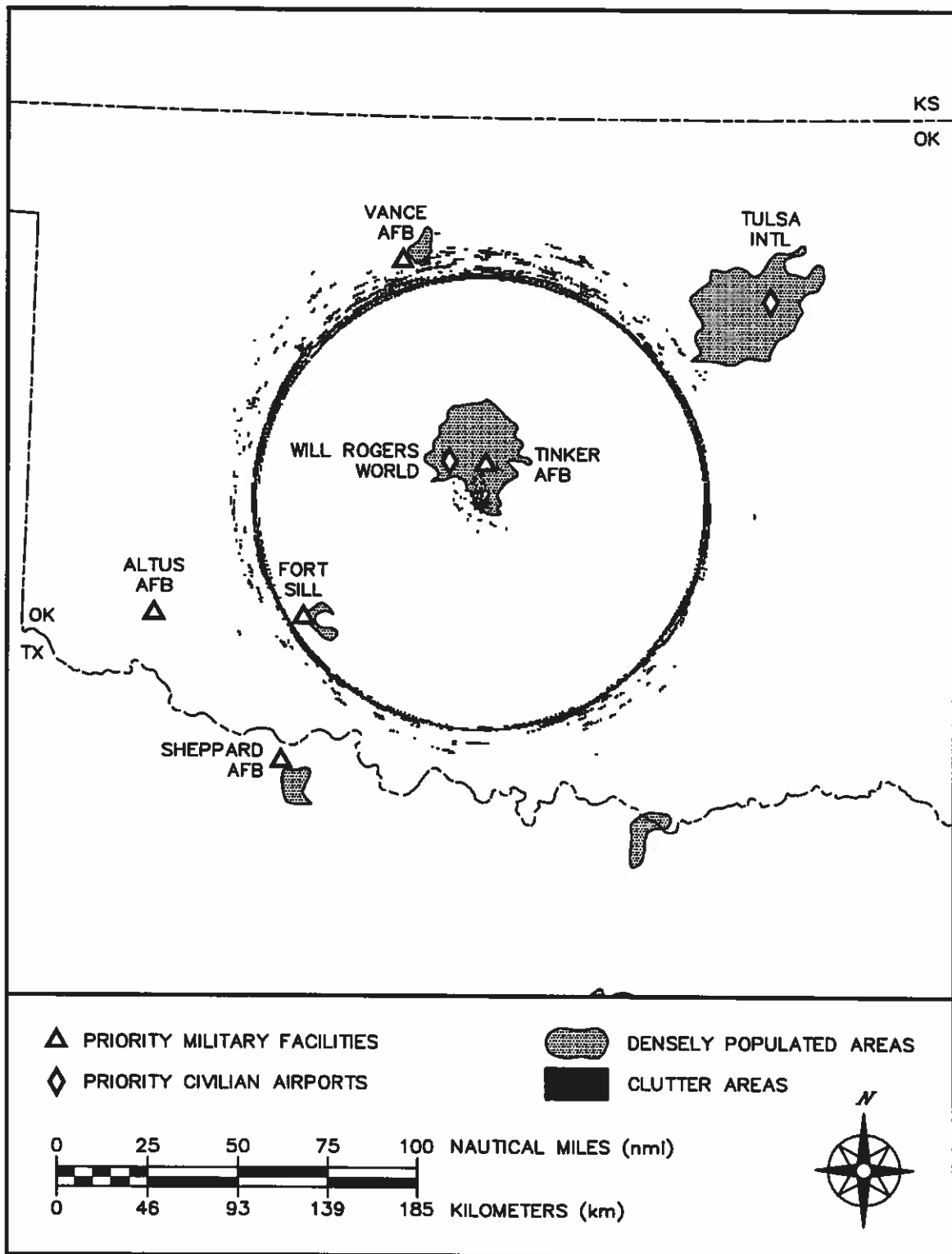


FIGURE 8 CLUTTER ESTIMATE, $> +29$ dBZ - PROPOSED OSF-3 SITE
 AT MAX WESTHEIMER AIRPARK, NORMAN, OKLAHOMA

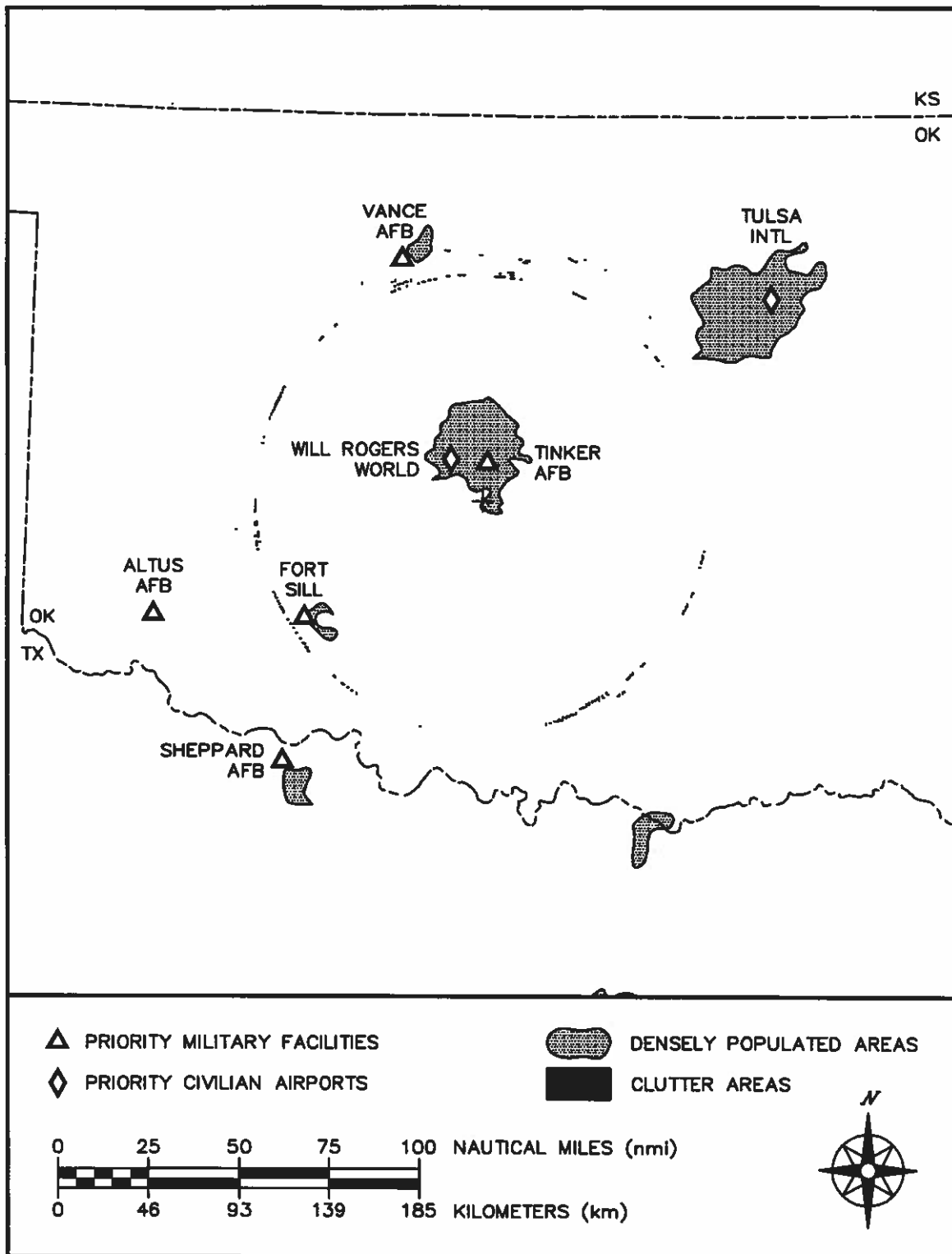


FIGURE 9 CLUTTER ESTIMATE, > +44 dBZ -- PROPOSED OSF-3 SITE
AT MAX WESTHEIMER AIRPARK, NORMAN, OKLAHOMA

areas, airports identified by the FAA as important, priority DoD facilities, and geographical areas over which clutter returns would degrade weather observations.

1. Pulse-repetition Frequency Effects

The pulse-repetition frequency (PRF) used determines the areas over which clutter obscures weather information. To obtain reflectivity data, NEXRAD would operate at low PRFs. For low PRF values, the unambiguous radar range is large (248 nmi) and the clutter area is centered, roughly, on the radar site (i.e., only near-in clutter is of concern). To obtain unambiguous radial velocity information, NEXRAD would also operate at high PRFs (up to approximately 1,300 Hz). For high PRF values, the interpulse interval is short, and the unambiguous radar range is limited to about 62 nmi; however, NEXRAD is required to provide Doppler measurements of weather out to 124 nmi. In this situation, low PRF (large unambiguous range) reflectivity data are used to unfold the high PRF velocity data and place them at the correct range. In the velocity-data channel operating at high PRFs, each clutter element would affect weather elements at more than one location. Therefore, Figures 7, 8, and 9 show a second, outer band of clutter (i.e., areas over which second-time-around weather echoes would be obscured by the clutter). The inner radius of this band could be as small as 62 nmi; the actual radius would depend on the PRF. Thus, by changing the PRF, the second clutter band could be moved (by about 10 percent) to improve the coverage over a particular area of interest.

2. Near-in Clutter

Figure 7 represents the estimated land clutter returns exceeding 0-dBZ weather returns for NEXRAD. Near-in clutter returns are estimated to extend to an average radial distance of 8 nmi. Figure 8 represents the estimated land clutter returns exceeding +29-dBZ weather returns for NEXRAD. Near-in clutter returns of this level are expected to extend to a radial distance of less than 6 nmi and from isolated structures and terrain features out to a range of 9 nmi. Figure 9 represents the estimated land clutter returns exceeding +44 dBZ weather returns for NEXRAD at the lowest elevation (+0.5 deg) and 0-dBZ weather returns for NEXRAD at higher elevations (+1.5 deg and higher). Near-in clutter returns of this level are expected to extend to a radial distance of less than 1 nmi and from isolated structures or terrain features out to a range of 7 nmi. An antenna elevation angle of 0.5 deg is assumed in all cases. The clutter estimates of Figures 7, 8, and 9 include 30 dB of clutter cancellation.

3. Second Clutter Band

The second clutter band (displayed in Figures 7, 8, and 9) would have the same intensity as the near-in clutter; however, weather-return intensity falls off as $1/R^2$, where R is the range from the radar to the weather. The second clutter band would, therefore, mask more intense weather than that masked by the near-in clutter. For the conditions

shown in Figure 7, weather returns weaker than +12 dBZ (corresponding to rain falling at less than 0.01 inch/hr) would generally be obscured toward the outer edge of the second clutter band. Toward the inner edge of the band, weather returns of higher intensity would be obscured. For the conditions shown in Figure 8, weather returns weaker than +50 dBZ (corresponding to rain falling at 2.0 inch/hr) would generally be obscured toward the outer edge of the second clutter band. Toward the inner edge of the band, weather returns of even higher intensity (typically 30 dB higher) would be obscured. The following areas of desired coverage are located where, depending on the PRF being used, weather observations could be affected by the second clutter band (see Figure 8).

Cities	Airports
None	None

4. Effects of Scan Scenario and Antenna Pattern

The clutter estimates presented above are for the lowest operational elevation angles of the NEXRAD antenna. Typical operational scan scenarios consist of azimuth scans at up to 14 elevation angles ranging from 0.5 to 19.5 deg. For elevation angles above 1.15 deg, the clutter returns would be reduced by approximately 50 dB* because only side lobes would illuminate the land. Under these conditions, clutter returns exceeding 0-dBZ weather returns should extend to a radius less than 2 nmi, and clutter returns exceeding +29-dBZ weather returns should be limited to distances less than 1 nmi. For the second clutter band, the decrease in clutter intensity would not lessen the masking of weather returns below approximately 11,000 ft, because the low-altitude weather would not be illuminated when the beam was elevated above 1.15 deg.

5. Summary

Within a radius of 62 nmi, a NEXRAD at the proposed OSF-3 site should reliably detect weather beyond 1 nmi without interference from near-in clutter returns. That is, near-in clutter should not obscure weather returns as severely as depicted in Figures 7 and 8, which were derived for an antenna elevation of 0.5 deg. The second clutter band would, however, obscure weather returns below approximately 11,000 ft in the areas shown in Figures 7 and 8.

*This value is derived from the NTR, which specifies that the first side lobe shall be at least 25 dB weaker than the main beam.

D. Anomalous Propagation

Anomalous propagation (AP) can exist in the Oklahoma City area at any time of the year. It most often occurs in the evening hours and is usually associated with temperature inversions. Anomalous propagation is most common toward the south and southwest. Anomalous propagation returns can be seen at ranges up to about 100 nmi. When atmospheric mixing occurs because of storm activity, or as the sun rises and heats the air, the inversions break down and the AP usually disappears. Radar weather observations are seldom affected by this phenomenon.

III RADAR TOWER HEIGHT

The availability and accuracy of weather data generated by the NEXRAD network will be affected by the clutter pattern at each site. Clutter patterns are affected by nearby obstructions surrounding the site and by the radar tower height. To obtain unambiguous Doppler spectrum measurements of weather echoes, NEXRAD will operate at high PRFs. The resulting range ambiguity will cause the near-in clutter pattern to be repeated, beginning as close as 62 nmi—depending on the PRF used. To minimize the intensity and extent of the clutter, considerable attention has been devoted to selecting sites having natural-terrain features that would act as clutter fences.*

There are six possible tower heights, ranging from 5 to 30 m in 5-m increments; the objective is to select the lowest practical height. An unnecessarily high tower increases the extent and intensity of clutter and increases construction and maintenance costs; if the tower is too low, the radar beam can be blocked.

Fresnel-zone clearance need not be considered when selecting the NEXRAD tower height. Radar energy transmission is unlike full-power transmission in a typical microwave line-of-sight (MLOS) communications link, where the direct ray from source to receiver should clear any intervening ridge by $0.6 F_1$ to $1.0 F_1$, or more, (where F_1 is the radius of the first Fresnel zone). For MLOS links, the area of the receiving antenna is very small compared with the area of the transmitted beam at the receiving end. The $0.6 F_1$ criterion is inappropriate for the radar beam because (1) weather targets are large and beam filling, (2) the $0.6 F_1$ criterion is derived on the basis of a uniform wavefront—whereas the NEXRAD beam is narrow, having most of the energy concentrated near the beam axis, and (3) the Fresnel-zone constraint applies to the lowest rays of the radar beam only, which represent a small fraction of the total beam energy.

Assuming that the minimum elevation angle of the main beam will usually be $+0.5$ deg and that the 3-dB beamwidth will be 1.0 deg, it is appropriate to position the center of the NEXRAD antenna to be level with the average height of nearby trees in the directions of interest. Emphasis is given to directions in which nearby cities are likely to produce undesirable levels of clutter. Within about 1,100 ft from the radar, before the NEXRAD beam begins to spread, no obstruction can be tolerated and a 1:1 margin must be allowed for tree growth. Beyond this distance, minor present or future blockage may be tolerable.

*Doviak, R. J., and D. S. Zrnic, "Siting of Doppler Weather Radars to Shield Ground Targets," *IEEE Transactions, Antennas and Propagation*, Vol. AP-33, No. 7, pp. 685-689 (July 1985).

For OSF-3, the height to the center of the antenna was selected to allow collocation with the KOUN WSR-88D.

A minimum 10-m separation in tower height with the 20-m KOUN WSR-88D is required to avoid mutual electromagnetic interference (EMI). Both a 5-m and 10-m tower could lead to a perception of RF radiation hazard (see Appendix B).

Therefore, a tower height of 30 m (corresponding to a height of 114 ft AGL to the center of the antenna) is recommended.

IV RADAR FREQUENCY ASSIGNMENT

Frequency assignment recommendations are based on information contained in ECAC's data base and on RF measurements taken by SRI International at the site. Measurement data were correlated with the ECAC data-base information. We assume that unseen radiators contained in ECAC's data base were still active but were not operating during the period of SRI's measurements. Two tentative frequency selections (a primary and a backup) are recommended on the basis of the following selection criteria:

- All recommended frequency assignments are in the 2.7- to 2.9-GHz band if possible; recommendations are in the 2.9- to 3.0-GHz band only when analysis shows that all available frequencies in the 2.7- to 2.9-GHz band would cause interference.
- Special consideration is given to selecting a frequency assignment that would minimize third harmonic reception by very large array (VLA) and very long baseline array (VLBA) radio-astronomy facilities.
- Special consideration is also given to selecting a frequency assignment that would not trigger the U.S. Coast Guard's radar transponder beacons (RACON) when these facilities are within 100 nmi.
- Neighboring NEXRADs are given different frequency assignment recommendations.

These criteria should result in frequency assignments that are scattered across the 2.7- to 2.9-GHz band.

As shown in Figure 10, the 2.7- to 2.9-GHz band in the Norman area is congested. The ordinate scale of Figure 10 represents maximum received pulse power through an antenna having a gain of 0 dBi. Three radiators (2723 MHz, -38.6 dBm; 2883 MHz, -24.0 dBm; 2887 MHz, -12.9 dBm) were seen that were not listed in the ECAC data base. After identifying nearby radiators and analyzing applicable frequency-distance curves,* a 2950-MHz primary frequency assignment is recommended (see Appendix B). Waveguide bandpass filters would be required in both the KOUN and the OSF-3 NEXRAD systems (see Appendix B). A frequency assignment in the 2700-MHz to 2750-MHz range has been recommended in an OSF document.†

*Sangston, Kevin J., and Daniel B. Schwartz, *Technical Characteristics and Frequency-Distance Relationships for NEXRAD and Other Systems in the 2,700-3,000-MHz Band*, Department of Defense Electromagnetic Compatibility Analysis Center, Annapolis, Maryland (February 1985).

†WSR-88D Operational Support Facility Redundant Radar Siting Consideration and Recommendations by Dale Sirmans and Paul Bontemp (July 5, 1994).

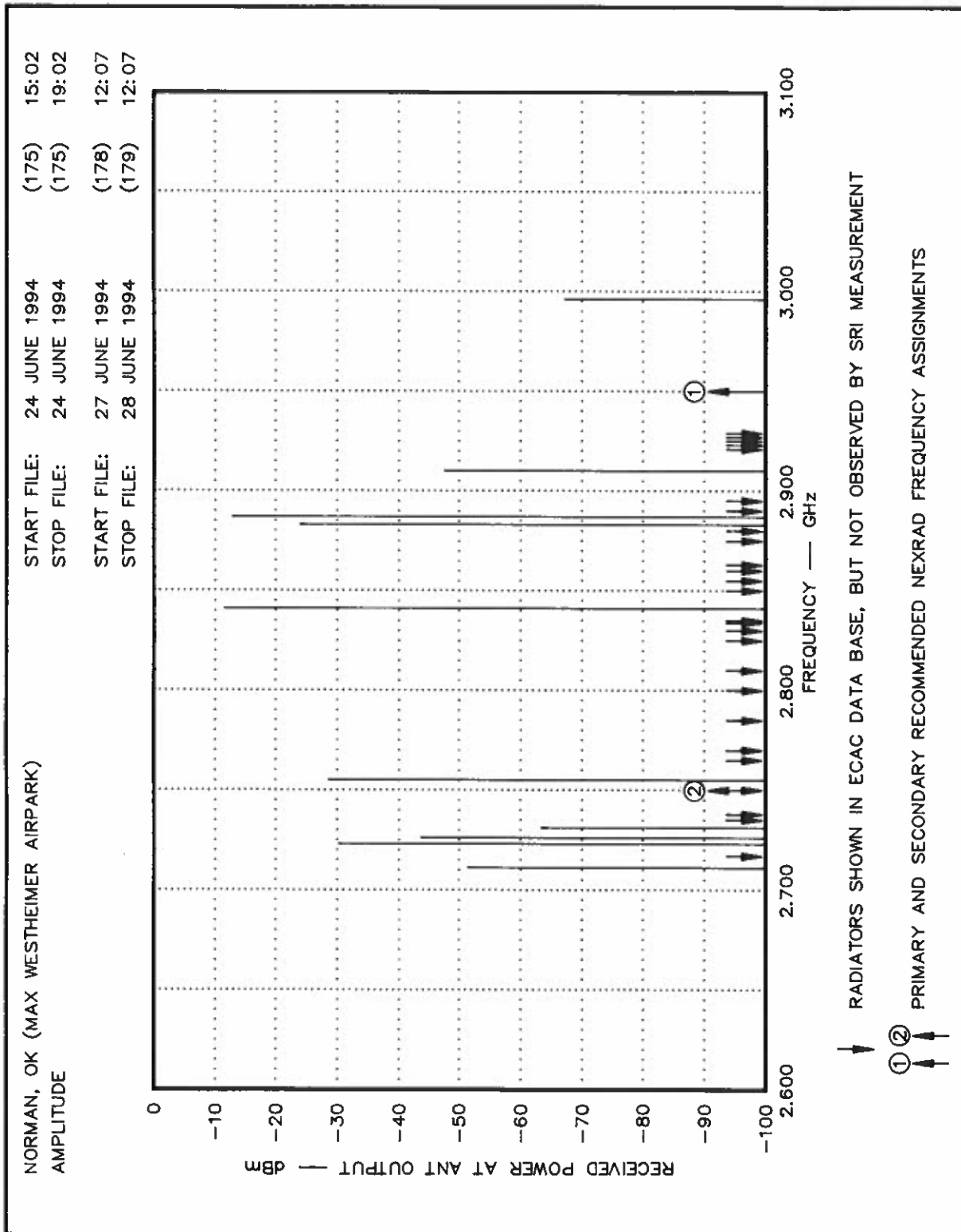


FIGURE 10 FREQUENCY ASSIGNMENT ANALYSIS — PROPOSED OSF-3 SITE AT MAX WESTHEIMER AIRPARK, NORMAN, OKLAHOMA

The frequency assignment recommendations have been forwarded to ECAC for detailed INR (interference-to-noise ratio) analysis and review with respect to existing frequency assignments in the 2.7- to 3.0-GHz frequency band.

V ROADS AND UTILITIES

A. Roads

1. Existing Conditions

Access to the proposed radar site is via Interstate Highway 35, Robinson Street, Halley Avenue, and Westheimer Drive. No steep grades or restricted bridges in the immediate area would limit access by HS-20-44 (alternative loading) vehicles.* Road maintenance of Interstate Highway 35 is performed by the Oklahoma Department of Transportation. Road maintenance of Robinson Street is performed by the City of Norman Department of Public Works. Road maintenance of Halley Avenue and Westheimer Drive is performed by the University of Oklahoma.

2. Future Needs

An approximately 60-ft long, 12-ft wide, site driveway will be constructed for access from Westheimer Drive to the radar facility.

B. Water Supply

1. Existing Conditions

An 8-inch water main is present along Westheimer Drive, approximately 80 ft from the site (Ken Jorgenson, personal communication, June 1994).

2. Future Needs

Water service is not required at the proposed radar site.

* *Standard Specifications for Highway Bridges*, Fourteenth Edition, American Association of State Highway and Transportation Officials, Washington, D.C. (1989).

C. Sewage Disposal

1. Existing Conditions

Sewer service is available from an 8-inch sewer approximately 110 ft north of the site (Ken Jorgenson, personal communication, August 1994).

2. Future Needs

Sewer service is not required at the proposed radar site.

D. Primary and Standby Electric Power

1. Existing Conditions

An aboveground, 12.5-kV, 3-phase power line, supplied by Oklahoma Gas & Electric Company and operated by the University of Oklahoma, runs along Westheimer Drive, adjacent to the radar site (Ken Jorgenson, personal communication, August 1994).

2. Future Needs

Three-phase power must be extended approximately 50 ft to the proposed site. The NWS plans a 200-A main, 208Y/120-V, 3-phase, 4-wire service for the radar.

The University of Oklahoma would provide and maintain primary wiring, secondary wiring, transformers, trenching, meter, and meter base. The NWS would provide and maintain service wiring. A six-month lead time would be necessary to provide service to the site (Ken Jorgenson, personal communication, August 1994).

A new 80-kW standby generator in a separate shelter is planned to serve the NEXRAD.

E. Fuel Storage

1. Existing Conditions

No fuel storage tanks are on the proposed site (Ken Jorgenson, personal communication, August 1994).

2. Future Needs

The 80-kW NEXRAD standby generator shelter contains built-in fuel storage.

F. Communications

1. Existing Conditions

Southwestern Telephone Company provides underground telephone service along Westheimer Drive, adjacent to the radar site. The cable originates at the central office, approximately 2 nmi away in Norman. There is fiber-optic capability available in the University of Oklahoma Research Park also (Kathy Lane, personal communication, August 1994).

2. Future Needs

One pair of commercial telephone lines is planned at the proposed radar site. Two sets of site plans will be required by Southwestern Bell Telephone Company upon service application. A lead time of approximately two months would be required to provide service to the proposed site (Kathy Lane, personal communication, August 1994).

VI ENVIRONMENTAL ASSESSMENT

This portion of the report considers site-specific environmental issues, both short-term and long-term. Short-term effects are those caused by construction activities; long-term effects are those resulting from NEXRAD operations. Methods to mitigate any short-term or long-term effects identified are outlined in Section D below. If a significant impact is expected in one of the following areas, a site-specific Environmental Impact Statement (EIS) could be required.

A. Potential Impact on Existing Environmental Conditions

The following paragraphs describe the potential short- and long-term effects of installing NEXRAD at the proposed site.

- (1) **Mineral Resources** The site is at Max Westheimer Airpark, a general aviation airport and research park. Oil and gas wells are located in the vicinity; the nearest active well is about 1,200 ft south of the site. Construction of NEXRAD at this location would not hinder operation of existing wells or prevent future hydrocarbon exploration. No impacts on the availability or use of mineral resources will result.
- (2) **Water Resources** Runoff from the site and vicinity flows into a concrete-lined drainage ditch along the north side of Westheimer Drive. Water in that ditch flows westward through several culverts under driveways connecting to Westheimer Drive and crosses under Westheimer Drive via two concrete culverts about 750 ft west of the site. South of Westheimer Drive, those culverts discharge into a brushy ditch which connects to Merkle Creek about 2,000 ft southwest of the site. Merkle Creek, an intermittent stream, flows southward and discharges into the Canadian River about 2.5 nmi southwest of the site.

The NEXRAD site is located in a mowed grassy field. Construction of NEXRAD and the access drive would result in removal of vegetative cover from about 0.15 acre, which would be covered with impervious surfaces including the NEXRAD tower foundation, shelter, and crushed rock. Because of the small amount of impervious surfaces created—less than one-sixth acre—the increase in surface runoff would be insignificant.

The construction and operation of the proposed facilities would not threaten subsurface water resources. Local regulations regarding the placement of fuel storage and other support facilities would be adhered to.

The proposed facilities would have no long-term effects on water resources.

- (3) **Hydrological Processes (Flooding, Erosion, and Sedimentation)** Max Westheimer Airpark is outside the area of detailed study by the Federal Emergency Management Agency (FEMA).^{*} However, the site is well drained by existing ditches and the base floodplain near Merkle Creek downstream of the Airpark is confined to the stream channel and adjacent areas. There are no indications of widespread flood hazards in the area. The site is not near a stream channel, and there appears to be a very low potential for flooding.

Soil at the site consists of Doolin-Urban Land-Pawhuska complex on 0 to 3 percent slopes. This soil is a deep, moderately well-drained silt loam. Runoff is slow, and the erosion hazard is high. The soil has a high clay content, causing low strength and high potential for shrink-swell behavior.[†] Standard measures will be implemented during construction to prevent excessive soil erosion and washing of soil into nearby ditches. After construction is complete, disturbed areas will be covered with erosion-resistant materials, primarily crushed rock, and long-term soil erosion should be minor. No long-term effects on these processes would be caused by the proposed facilities.

- (4) **Air Quality** Construction equipment would create diesel exhaust during the four- to six-month construction period. Because of the limited amount of equipment required to develop the facilities and the short construction period, diesel exhaust would not have a significant short- or long-term effect on air quality. Construction activities could raise dust, which could have a short-term impact on local air quality and visibility. Dust-suppression measures, such as daily watering of exposed soil during the construction period, will be implemented to minimize dust generation. The direct, long-term effect on air quality by exhaust from the standby diesel power generator, which would operate infrequently, is negligible.

- (5) **Flora and Fauna** The U.S. Fish and Wildlife Service (USFWS) and the Oklahoma Department of Wildlife Conservation were contacted concerning possible effects on listed species. No endangered or threatened species are known to occur at the site or vicinity, and no adverse effects on listed species are expected. The USFWS and the State of Oklahoma concur with that determination (see Appendix B). NEXRAD construction would remove vegetation from about 0.15 acre in a developed area; any loss of wildlife habitat would be insignificant.

The radar should have no adverse effects on migratory or indigenous birds. Any heating effects (even in the main beam) are expected to be negligible. The proposed radar would be similar to the existing KOUN NEXRAD, located at

^{*}FEMA Flood Insurance Rate Map (FIRM), City of Norman, Cleveland County, Oklahoma, Community-Panel Number 400046-0015B (November 1, 1979).

[†]Soil Conservation Service, *Soil Survey of Cleveland County, Oklahoma, Area* (April 1987).

Max Westheimer Airpark about 900 ft southwest of the proposed site. The existing radar, in operation since 1988, has not resulted in significant hazards to birds. Bird strikes are possible, as with any structure; however, because the NEXRAD tower and radome are relatively short (less than 150 ft AGL) and highly visible, significant bird losses due to tower collisions are not anticipated.

A line of small planted trees occurs in the grassy area containing the site. Roughly two to four trees would be removed for NEXRAD construction.

- (6) **Environmentally Sensitive Areas (Wetlands and Floodplains)** No environmentally sensitive areas, such as wetlands and floodplains, have been identified at the site. The nearest wetland is a brushy area at the upper reaches of Merkle Creek, about 750 ft west of the site. No adverse effects on those wetlands are expected.

The site is not within a 100-year floodplain.

- (7) **Noise** Noise from construction activities and construction equipment traffic would have a short-term impact on nearby airport facilities and office buildings. To mitigate that impact, construction will occur only during normal working hours.
- (8) **Aesthetics** The visual quality of the area is characterized by airport facilities. The primary aesthetic impact would be the view of the NEXRAD tower and radome. This is not considered unusual in this airport environment. No adverse aesthetic impact is anticipated.
- (9) **Transportation** Traffic in the immediate vicinity of the proposed radar site is moderate. Construction activities would create traffic during the four- to six-month construction period; however, this would be a minor short-term effect. The NEXRAD facility would be unmanned and would not significantly increase long-term traffic through residential areas and should have no long-term impact. Maintenance would be performed by personnel from the NEXRAD OSF administrative building, about 300 ft from the site, and would not generate vehicle traffic.
- (10) **Man-made Facilities** The nearest man-made facilities to the site are the OSF administration building, about 300 ft to the south; a small garage, about 300 ft to the northwest; two abandoned former military buildings, about 300 ft to the west and northwest; and the National Severe Storms Laboratory, about 500 ft to the west-southwest. A YMCA building is about 500 ft to the north. The KOUN NEXRAD is about 900 ft to the southwest and the Norman Doppler Radar is about 1,200 ft to the southwest. The NEXRAD tower, equipment shelter, and diesel generator would not be unusual in this airport setting. In fact, the proposed NEXRAD would be very similar in appearance and operation characteristics to existing radars and water towers at Max Westheimer Airpark.

The proposed site is within the University of Oklahoma Research Park, which is a portion of Max Westheimer Airpark. Existing and planned uses at the research park include research, academic, office, and industrial uses. NEXRAD would be compatible with those uses.*

No adverse effects on the surrounding man-made facilities will result.

- (11) **Public Services** The following public services are required for the proposed facilities.

- Fire department
- Police department
- Electrical power
- Telephone.

The demand on these services from the construction and use of the proposed facilities would be minor.

- (12) **Recreational Resources** Other than aviation-related activities and the YMCA facility, no established recreational resources are in the immediate area. The proposed facilities would have no long-term effects on recreational resources.

- (13) **Cultural Resources** The Oklahoma State Historic Preservation Office (SHPO, a.k.a. Oklahoma Historical Society) and the Oklahoma Archaeological Survey were contacted about the proposed action. No historical or archaeological resources listed or eligible for listing on the National Register of Historic Places are known to occur at the site or vicinity. The potential for artifacts to be uncovered at the proposed site is low (see Appendix B). No impacts on cultural resources are expected. However, in the unlikely event that potentially significant archaeological materials are uncovered during construction, the Oklahoma Archaeological Survey will be notified so that they can evaluate the significance of the find.

- (14) **Community Relations** Previous development in the area has not been opposed by local area residents. The existing radars at the Airpark have not generated an adverse reaction from the public.

- (15) **Hazardous Materials** Except for fuel for the standby generator, the proposed facilities will contain no known hazardous materials.

From the information available on current ownership and use of the site, as well as from a visual inspection of the premises, no indication of soil or surface water contamination by toxic substances was detected.

*George Hargett, Administrator, University Research Park, University of Oklahoma, personal communication, May 17, 1994.

B. Impact from Radio-Frequency (RF) Radiation

In selecting the operating frequency for NEXRAD, we considered all existing radars and communications equipment nearby, as well as the intended adjacent NEXRADs.

1. Human Health

The effects of microwave radiation are often confused with the effects of higher frequency radiation such as ultraviolet light, X-rays, and gamma rays. These higher frequency forms of radiation are called ionizing radiation because they can displace electrons from molecules in the human body, causing permanent damage. Microwave radiation (such as will be emitted by NEXRAD) is called nonionizing radiation because it cannot displace electrons. However, nonionizing radiation can cause heating of body tissues if the amount of energy absorbed by the tissue exceeds the ability of the body's thermoregulatory system to dissipate heat.

The heating effects of microwaves can be understood by analogy to visible light. Both visible light and microwaves can increase the vibration rate of molecules, an effect that at a macro-scale is called heating. Excessive heating can be dangerous, as exemplified by sunstroke; hence, the rate of heating is critical. Many adverse biological effects have been shown to result from exposure to RF radiation that exceeds the body's ability to dissipate heat. A limited number of biological effects have also been demonstrated from exposure to non-thermal levels of RF radiation, but those effects have not been demonstrated to be harmful. Microwaves heat most effectively those objects that have sizes, shapes, or compositions that trap the radiation through resonance or absorption. Both resonance and absorption rates are very sensitive to the frequency used. The NEXRAD system will use frequencies that are inefficiently absorbed by humans and wildlife, and consequently cause little heating. The amount of energy absorbed by humans exposed to NEXRAD RF radiation is well below the ability of the body's thermoregulatory system to dissipate heat. Thus, exposure to NEXRAD RF radiation will not cause heating of body tissues.

When a NEXRAD antenna is at its lowest practical height (i.e., on a 5-m tower), the highest RF radiation average power density to which the general public could be exposed is expected to be 0.003 mW/cm^2 —more than a factor of 600 lower than the current U.S. standard ($1.8 \text{ to } 2.0 \text{ mW/cm}^2$) for the NEXRAD operating frequencies.* With the NEXRAD antenna on the proposed 30-m tower, the highest RF radiation average power density to which the general public could be exposed at the base of the tower is expected to be 0.0003 mW/cm^2 —a factor of 6,000 lower than the current U.S. standard. Significantly higher exposure levels will be possible only if some combination of radar tower height and terrain elevation or structure height causes illumination by the main beam. However, the highest average power density in the main beam that someone on

* IEEE C95.1-1991, "IEEE Standard for Safety Levels with Respect to Human Exposure to Radio Frequency Electromagnetic Fields, 3 kHz to 300 GHz," Institute of Electrical and Electronics Engineers, Inc., New York City, New York (1991).

or very near the radome could experience will be approximately 0.6 mW/cm^2 , about one third the U.S. standard. The radiation intensity of bright sunlight is more than 100 mW/cm^2 —that is, more than 50 times as large as the U.S. standard for microwave radiation, and 333,300 times as large as the 0.0003 mW/cm^2 calculated for NEXRAD at this tower height. This comparison makes it clear that the body heating effect of NEXRAD will be negligible.

The combined RF radiation levels caused by the KOUN and OSF-3 NEXRADs, at ground level near the locations of the NEXRADs, would be 2,900 times less than the maximum permissible exposure level set by ANSI/IEEE (1992) guidelines. There are no television or radio transmitters close enough to add significantly to the RF radiation level. No adverse health effects will result from exposure to RF radiation from the two NEXRADs and other RF radiation sources.

2. Radio and Television Interference

High-powered radars can interfere with radio and television reception in their immediate vicinity. The emission spectrum of NEXRAD is expected to be superior to that of the WSR-57 and the WSR-74S weather radars currently operating at other sites. Because radio and television interference has not been attributed to the existing KOUN NEXRAD at Max Westheimer Airpark, problems are not expected from the improved NEXRAD emission spectrum. Tests conducted at the NEXRAD Operational Support Facility in Norman, Oklahoma, by the Department of Defense ECAC, demonstrated that interference with television and radio reception by the NEXRAD signal is very unlikely.*

3. Blasting Operations

Blasting caps can be detonated by electromagnetic energy. Calculations based on an Air Force standard[†] indicate that using electric blasting caps at distances beyond about 900 ft is a safe practice, even in the main beam of the NEXRAD, where illumination is most intense (a worst-case situation).

Because the main beam would not be directed downward far enough to illuminate objects close to the ground (within 900 ft), blasting operations in the vicinity would not be jeopardized by NEXRAD operation.

*Kenney, D. and A. Koppenhaver, *Effects of WSR-88D on Residential Television Reception*, Electromagnetic Compatibility Analysis Center, ECAC-CR-89-092 (1989).

[†]*Explosives Safety Standards*, U.S. Air Force Regulation 127-100(C1), 31 March 1978 (Change 1, 18 June 1979) (revised draft obtained 27 July 1982).

4. Liquid Fuels

Calculations based on another Air Force standard*—dealing with the transfer of liquid fuels—indicate that fuel-handling activities would be safe, even in the main beam of the NEXRAD. Thus, there would be no apparent danger from RF radiation.

5. Interference with KOUN NEXRAD

Electromagnetic interference with the existing KOUN NEXRAD at Max Westheimer Airpark will be prevented by selecting a frequency for the OSF-3 NEXRAD that provides adequate frequency separation, providing adequate filtering, and by installing the OSF-3 NEXRAD on a different height tower than the existing KOUN NEXRAD. The existing KOUN NEXRAD operates at a frequency of 2995 MHz; therefore, a frequency assignment of 2950 MHz is recommended for the OSF-3 NEXRAD. The existing KOUN NEXRAD is mounted on a 20-m tower. The OSF-3 NEXRAD will be mounted on a 30-m tower and will not transmit below 0°.

6. Interference with Oklahoma City NEXRAD MLOS Tower

The upper MLOS antenna that serves as a data link between the OSF facilities and the Oklahoma City NEXRAD at Twin Lakes is likely to receive interfering signals from the OSF-3 NEXRAD more than 20 dB stronger than those now impinging on either of these MLOS antennas. If interference occurs, it can be suppressed by additional filtering, or by the addition of a metal screen to deflect incoming radiation (see Appendix B).

C. Summary of Potentially Significant Impacts

No potentially long-term environmental effects are expected.

Potential short-term effects are as follows:

- Soil sedimentation, erosion, and runoff during the construction period
- Dust created during the construction period
- Noise generated by construction equipment
- Construction traffic on the airpark
- Uncovering archaeological resources during construction.

None of the effects is considered to be significant, if mitigated by the measures listed below.

**Electromagnetic Radiation Hazards*, U. S. Air Force Communications Service (E-1 Standard), Technical Manual T.O. 31Z-10-4, 1 August 1966 (Change 2, 1 June 1971).

D. Methods to Mitigate Impacts

Sedimentation and erosion should be alleviated by temporary control devices during construction, such as straw bales in drainage ways and grading slopes for runoff retention, and by proper restoration of the site after construction is completed.

Construction activities should include dust control procedures, such as periodic watering, to minimize the impact on air quality and the surrounding airport area.

To control noise, work should be performed during normal working hours, and equipment should be shut off when not in use.

To limit the effects of short-term, construction-related traffic, the NEXRAD construction period should be kept to a minimum. Specific temporary truck routes could be established to reduce construction vehicle traffic near residential areas.

If possibly significant archaeological resources are uncovered during construction, earth movement should be suspended, and the Oklahoma Archaeological Survey should be notified.

E. Key Environmental Impact Issues

No significant impacts on the quality of the human environment will result if the mitigation measures listed above are implemented. A site-specific EIS is not considered necessary at this time. Preparation of a Finding of No Significant Impact (FONSI) is warranted.

VII FEDERAL AVIATION REGULATIONS COMPLIANCE

SRI has submitted FAA Form 7460-1, "Notice of Proposed Construction or Alteration," for the proposed NEXRAD tower at Max Westheimer Airpark (see Appendix B). The FAA will undertake a detailed study based on this application, and may impose certain restrictions or requirements that it deems necessary to ensure flight safety.

Table 2 presents an analysis of the interference potential between the proposed NEXRAD facilities and the existing and proposed FAA facilities at or near Max Westheimer Airpark. Most areas of potential obstruction listed in this table were derived from FAA Handbook 7400.2D.*

A. FAR, Part 77 Requirements†

Figure 11 presents an analysis of the proposed NEXRAD tower in relation to the requirements of FAR, Part 77. The approximate ground elevation is drawn from the proposed NEXRAD tower to the Airport Reference Point. As can be seen, the NEXRAD tower would be within the height limitations established by the regulations. A preliminary TERPS evaluation indicates that the existing controlling obstacle height is 1,330 ft MSL. Therefore, a foundation grade level increase of no more than 7 ft can be used.

B. Obstruction Marking

No marking or lighting is required by FAA Advisory Circular 70/7460-1H for this proposed radar tower. However, the FAA may require lighting and/or marking as a condition of approval after reviewing FAA Form 7460-1.

C. Summary

In summary, this analysis has revealed no issues that might affect FAA approval.

* *Procedures for Handling Airspace Matters*, Handbook 7400.2D, Federal Aviation Administration, U.S. Department of Transportation (September 16, 1993).

† "Obstruction Requirements," Part 77, Subsection C, *Objects Affecting Navigable Airspace*, Federal Aviation Regulations, Department of Transportation (January 1975).

Table 2

**ANALYSIS OF INTERFERENCE POTENTIAL FROM NEXRAD FACILITIES TO
FAA AND AIRPORT FACILITIES—PROPOSED NEXRAD TOWER AT MAX WESTHEIMER AIRPARK**

FAA Facility	Distance to NEXRAD Facilities	Area of Potential Obstruction	Remarks
Closest Runway RWY 03-21	3350' to CL	See FAR, Part 77	Acceptable (see Figure 11)
ILS Localizer RWY 03	4200'/50°	± 35° of runway centerline and 500' from the localizer	Acceptable
ILS Localizer Future RWY 35*	9400'/42°	± 35° of runway centerline and 500' from the localizer	Acceptable
ILS Glide Slope Future 35	6650' to CL 150' FWD	Up to 3000' forward of antenna and within 1000' of runway centerline	Acceptable
VORTAC	10 nmi	Within 1500' radius and beneath a 1.2° imaginary surface	Acceptable
ASDE	N/A	Within 1500' radius and no shadowing of important ground-control points	N/A
ARSR	N/A	Within 3000' radius and no shadowing of ARSR beam	N/A
ASR 9 OKC	11 nmi	Within 3000' radius and no shadowing of important air-control points	Acceptable
ATCBI	11 nmi	Within 3000' radius and no horizontal reflections of beam pattern	Acceptable
DME	10 nmi	Within 300' radius	Acceptable
VHF/DF	N/A	Standard type—within 1500' radius Doppler type—within 3000' radius	N/A
NDB	5500'	Within 300' radius	Acceptable
Communications Facilities	2500'	Within 1500' radius	Acceptable
Control Tower	2500'	View of final approach and surface movement	Unobstructed view

* Per telephone conversation with Pauline at airport (405) 325-7233.

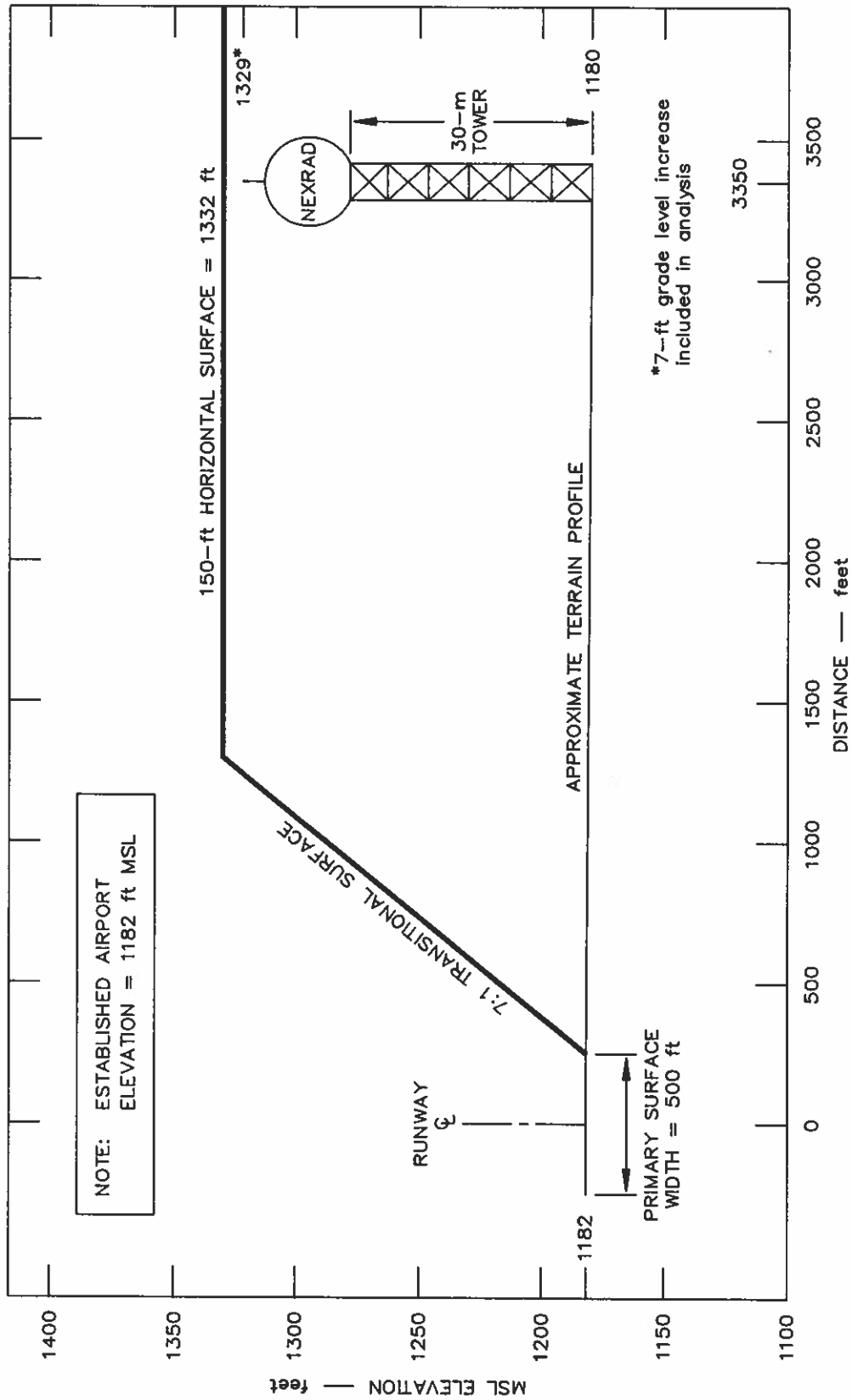


FIGURE 11 FAR, PART 77 ANALYSIS FOR RUNWAY 03/21 -- PROPOSED RADAR TOWER AT MAX WESTHEIMER AIRPARK, NORMAN, OKLAHOMA

VIII LONG LEAD-TIME ITEMS

This section provides a preliminary assessment of the procedures and lead times necessary for activities leading up to the start of construction. The federal government's relationship with local governments with respect to the application of local zoning laws and laws relating to the development of real property is set forth in Section 6 of the Public Buildings Amendments of 1988 (P.L. 100-678). (See Appendix B.) The NWS plans to follow the procedural requirements of that act.

The following information, which applies to non-federal developers, is provided to assist the NWS in its consultations with local officials.

A. Land-use Approval

The proposed radar site is owned by the University of Oklahoma and, therefore, is not classified in a zoning category. The NEXRAD tower would be an allowable use in this area.

The proposed site area has a minimum lot size of 1 acre, a 30-ft minimum front setback to improvement, and a 50-ft minimum setback to a building. All structures must be at least 25 ft from side- and rear-lot property lines (George Hargett, personal communication, August 1994).

B. Building Permits

Building plans need not be submitted or reviewed by the City of Norman Code Enforcement Department as the proposed site would be on University of Oklahoma property (Norma Holley, personal communication, August 1994). Three sets of site plans should be sent to the University of Oklahoma for review showing site layout, utility requirements, and structure elevations (Ken Jorgenson, personal communication, August 1994).

C. Power Service

A lead time of approximately six months would be required to provide the necessary power service to the proposed site (Ken Jorgenson, personal communication, August 1994).

D. Telephone Service

A lead time of approximately two months would be required to provide the necessary telephone service to the proposed site (Kathy Lane, personal communication, August 1994).

Appendix A

**ELECTROMAGNETIC COMPATIBILITY ANALYSIS CENTER'S
FREQUENCY ASSIGNMENT ANALYSES**

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Appendix B

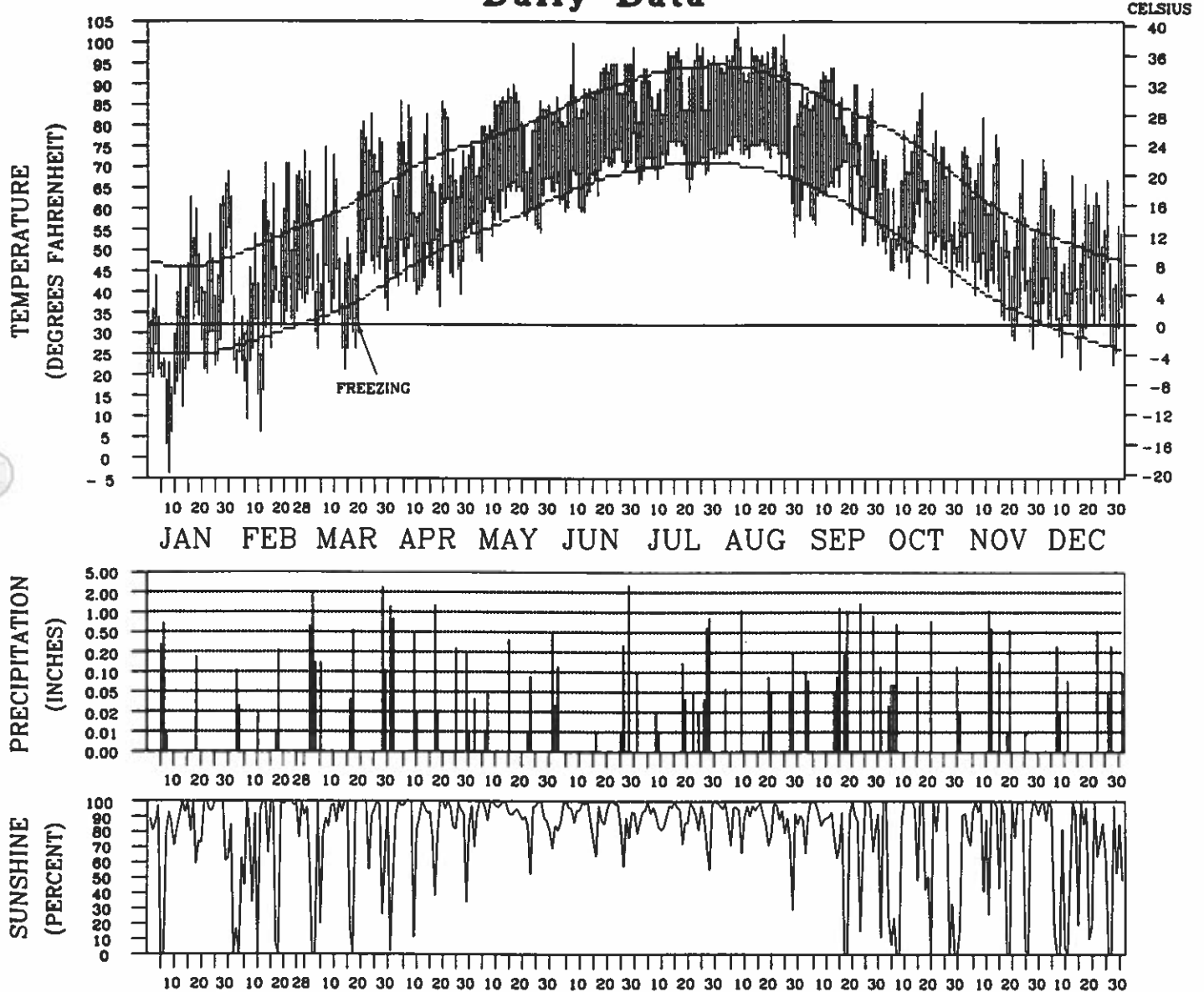
CORRESPONDENCE AND BACKGROUND INFORMATION

1988 LOCAL CLIMATOLOGICAL DATA ANNUAL SUMMARY WITH COMPARATIVE DATA

OKLAHOMA CITY,
OKLAHOMA



Daily Data



TEMPERATURE DEPICTS NORMAL MAXIMUM, NORMAL MINIMUM AND ACTUAL DAILY HIGH AND LOW VALUES (FAHRENHEIT)
PRECIPITATION IS MEASURED IN INCHES. SCALE IS NON-LINEAR
SUNSHINE IS PERCENT OF THE POSSIBLE SUNSHINE

I CERTIFY THAT THIS IS AN OFFICIAL PUBLICATION OF THE NATIONAL OCEANIC AND ATMOSPHERIC ADMINISTRATION, AND IS COMPILED FROM RECORDS ON FILE AT THE NATIONAL CLIMATIC DATA CENTER, ASHEVILLE, NORTH CAROLINA, 28801

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NATIONAL
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METEOROLOGICAL DATA FOR 1988

OKLAHOMA CITY, OKLAHOMA

LATITUDE: 35°24' N LONGITUDE: 97°36' W ELEVATION: FT. GRND 1285 BARO 1283 TIME ZONE: CENTRAL MBAN: 13967

	JAN	FEB	MAR	APR	MAY	JUNE	JULY	AUG	SEP	OCT	NOV	DEC	YEAR
TEMPERATURE °F:													
Averages													
-Daily Maximum	43.6	51.7	60.7	71.2	81.3	89.8	92.3	94.2	84.7	68.5	62.4	54.7	71.3
-Daily Minimum	24.7	28.8	38.3	46.6	59.2	66.9	70.9	71.4	62.2	50.0	40.0	33.1	49.3
-Monthly	34.2	40.3	49.5	58.9	70.3	78.4	81.6	82.8	73.5	59.3	51.2	43.9	60.3
-Monthly Dewpt.	23.8	25.4	32.8	42.5	53.5	60.9	66.0	65.0	58.9	46.9	37.1	29.1	45.2
Extremes													
-Highest	69	74	83	86	90	100	100	104	94	88	82	72	104
-Date	30	28	24	20	17	8	24	8	13	17	9	2	AUG 8
-Lowest	-4	6	21	36	47	59	64	53	51	41	26	21	-4
-Date	8	11	14	19	5	12	21	29	30	28	28	16	JAN 8
DEGREE DAYS BASE 65 °F:													
Heating	948	712	473	204	14	0	0	0	8	196	408	644	3607
Cooling	0	0	1	29	186	410	525	558	270	25	1	0	2005
% OF POSSIBLE SUNSHINE	82	71	73	85	91	90	89	87	79	59	75	59	79
AVG. SKY COVER (tenths)													
Sunrise - Sunset	5.1	5.0	4.6	5.2	5.3	4.9	5.4	4.5	4.2	5.0	3.7	5.0	4.8
Midnight - Midnight	5.1	4.4	4.3	4.7	4.9	4.4	4.7	4.3	4.1	4.5	3.1	4.3	4.4
NUMBER OF DAYS:													
Sunrise to Sunset													
-Clear	12	13	14	10	10	12	9	13	14	15	18	13	153
-Partly Cloudy	10	6	7	11	11	9	15	13	8	3	6	7	106
-Cloudy	9	10	10	9	10	9	7	5	8	13	6	11	107
Precipitation													
.01 inches or more	4	5	9	8	7	6	10	7	9	9	6	7	87
Snow, Ice pellets													
1.0 inches or more	2	0	0	0	0	0	0	0	0	0	0	0	2
Thunderstorms	1	0	4	5	5	6	7	6	8	3	3	2	50
Heavy Fog, visibility													
1/4 mile or less	0	1	1	0	0	0	0	1	0	2	3	2	10
Temperature °F													
-Maximum													
90° and above	0	0	0	0	1	13	21	27	9	0	0	0	71
32° and below	6	4	0	0	0	0	0	0	0	0	0	0	10
-Minimum													
32° and below	23	16	9	0	0	0	0	0	0	0	6	16	70
0° and below	1	0	0	0	0	0	0	0	0	0	0	0	1
AVG. STATION PRESS. (mb)	975.3	974.4	969.9	967.2	967.5	969.4	970.0	968.5	969.2	973.4	967.5	974.8	970.5
RELATIVE HUMIDITY (%)													
Hour 00	73	65	64	67	67	68	71	67	73	75	72	65	69
Hour 06 (Local Time)	80	72	72	76	79	78	84	81	82	80	76	72	78
Hour 12	58	53	50	48	49	44	52	45	52	58	50	51	51
Hour 18	63	50	49	43	43	44	47	45	53	62	56	53	51
PRECIPITATION (inches):													
Water Equivalent													
-Total	1.24	0.41	7.85	3.19	1.07	3.59	1.92	1.60	5.19	2.04	2.45	1.39	31.94
-Greatest (24 hrs)	0.87	0.24	2.95	1.86	0.48	3.10	0.87	1.13	1.47	0.79	1.70	0.54	3.10
-Date	5-6	18	28-29	PM-1	31	28	28	9	23	20	11-12	22	JUN 28
Snow, Ice pellets													
-Total	12.1	0.2	0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.6	2.0	15.8
-Greatest (24 hrs)	8.9	0.2	0.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.6	1.0	8.9
-Date	5-6	2	3								19-20	7-8	JAN 5-6
WIND:													
Resultant													
-Direction (!!!)	243	354	093	106	184	163	171	158	165	120	221	178	169
-Speed (mph)	1.7	3.7	1.0	0.2	5.4	6.5	5.7	6.1	3.4	3.1	2.7	1.9	2.5
Average Speed (mph)	13.7	13.3	14.7	13.2	12.9	10.4	11.1	10.1	10.9	10.9	13.9	12.0	12.3
Fastest Obs. 1 Min.													
-Direction (!!!)	19	35	12	26	17	13	32	27	28	02	26	17	26
-Speed (mph)	32	36	39	33	30	24	31	35	31	28	46	31	46
-Date	29	10	1	2	29	27	1	9	23	27	15	26	NOV 15
Peak Gust													
-Direction (!!!)	S	N	SE	N	S	NW	W	W	W	S	W	NW	W
-Speed (mph)	46	51	51	40	44	37	48	51	49	39	63	40	63
-Date	29	10	1	5	29	30	28	9	23	14	15	26	NOV 15

!!! See Reference Notes on Page 6B
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NORMALS, MEANS, AND EXTREMES

OKLAHOMA CITY, OKLAHOMA

LATITUDE: 35°24'N		LONGITUDE: 97°36'W		ELEVATION: FT. GRND 1285 BARD 1283		TIME ZONE: CENTRAL		WBAN: 13967						
	(in)	JAN	FEB	MAR	APR	MAY	JUNE	JULY	AUG	SEP	OCT	NOV	DEC	YEAR
TEMPERATURE °F:														
Normals														
-Daily Maximum		46.6	52.2	61.0	71.7	79.0	87.6	93.5	92.8	84.7	74.3	59.9	50.7	71.2
-Daily Minimum		25.2	29.4	37.1	48.6	57.7	66.3	70.6	69.4	61.9	50.2	37.6	29.1	48.6
-Monthly		35.9	40.8	49.1	60.2	68.4	77.0	82.1	81.1	73.3	62.3	48.8	39.9	59.9
Extremes														
-Record Highest	35	80	84	93	100	104	105	109	110	102	96	87	86	110
-Year		1986	1981	1967	1972	1985	1980	1986	1980	1985	1972	1980	1955	AUG 1980
-Record Lowest	35	-4	-3	3	20	37	47	53	51	37	22	11	-3	-4
-Year		1988	1979	1960	1957	1981	1954	1971	1956	1985	1957	1959	1983	JAN 1988
NORMAL DEGREE DAYS:														
Heating (base 65°F)		902	678	506	184	41	0	0	0	15	145	486	778	3735
Cooling (base 65°F)		0	0	13	40	147	360	530	499	264	61	0	0	1914
% OF POSSIBLE SUNSHINE	34	61	61	65	68	68	75	80	80	73	69	61	59	68
MEAN SKY COVER (tenths)														
Sunrise - Sunset	40	5.8	5.8	5.8	5.7	5.8	5.0	4.4	4.3	4.7	4.6	5.1	5.5	5.2
MEAN NUMBER OF DAYS:														
Sunrise to Sunset														
-Clear	40	10.5	9.0	9.8	9.6	8.8	10.8	14.5	14.7	13.4	14.2	12.1	11.6	139.0
-Partly Cloudy	40	6.2	6.9	7.9	7.9	10.0	10.5	9.5	9.8	8.0	6.8	6.7	6.0	96.2
-Cloudy	40	14.3	12.3	13.3	12.5	12.2	8.7	7.0	6.5	8.6	10.1	11.3	13.4	130.1
Precipitation														
.01 inches or more	49	5.4	6.4	7.1	7.7	10.0	8.5	6.4	6.4	7.0	6.4	5.2	5.3	81.9
Snow, ice pellets														
1.0 inches or more	49	0.9	1.0	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.6	3.1
Thunderstorms	49	0.5	1.4	3.0	5.4	9.0	8.6	6.1	6.2	4.8	3.1	1.3	0.6	50.0
Heavy Fog Visibility														
1/4 mile or less	40	3.8	3.2	1.9	0.9	0.6	0.3	0.3	0.3	0.8	1.7	2.3	3.4	19.4
Temperature °F														
-Maximum														
90° and above	23	0.0	0.0	0.2	0.4	1.7	11.6	22.7	23.0	8.9	0.8	0.0	0.0	69.3
32° and below	23	5.8	2.7	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	2.3	10.9
-Minimum														
32° and below	23	23.4	17.0	7.8	1.0	0.0	0.0	0.0	0.0	0.0	0.5	8.5	20.2	78.4
0° and below	23	0.4	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.7
AVG. STATION PRESS. (mb)	16	973.6	971.7	967.8	967.7	966.6	968.0	969.5	969.6	970.2	971.5	971.2	972.6	970.0
RELATIVE HUMIDITY (%)														
Hour 00	23	73	72	68	69	76	76	70	70	75	72	74	73	72
Hour 06	23	78	78	75	77	83	83	79	79	83	79	79	78	79
Hour 12 (Local Time)	23	59	58	52	51	57	56	49	49	54	53	56	58	54
Hour 18	23	60	55	48	49	54	53	45	46	53	55	60	62	53
PRECIPITATION (inches):														
Water Equivalent														
-Normal		0.96	1.29	2.07	2.91	5.50	3.87	3.04	2.40	3.41	2.71	1.53	1.20	30.89
-Maximum Monthly	49	5.68	4.05	7.85	10.78	12.07	9.94	8.44	6.77	9.64	13.18	5.46	8.14	13.18
-Year		1949	1987	1988	1947	1982	1979	1959	1966	1970	1983	1964	1984	OCT 1983
-Minimum Monthly	49	0.00	T	T	0.41	0.33	0.63	T	0.25	T	T	T	0.03	0.00
-Year		1985	1947	1940	1987	1942	1952	1983	1978	1948	1958	1949	1955	JAN 1985
-Maximum in 24 hrs	49	3.10	2.21	3.44	3.80	5.63	4.55	5.75	3.34	7.68	8.95	2.21	2.55	8.95
-Year		1982	1978	1944	1970	1970	1985	1981	1966	1970	1983	1986	1984	OCT 1983
Snow, ice pellets														
-Maximum Monthly	49	17.3	12.0	13.9	0.7	0.0	0.0	0.0	0.0	0.0	T	7.5	8.3	17.3
-Year		1949	1978	1968	1957						1967	1972	1987	JAN 1949
-Maximum in 24 hrs	49	8.9	6.5	8.4	0.7	0.0	0.0	0.0	0.0	0.0	T	5.5	8.3	8.9
-Year		1988	1986	1948	1957						1967	1972	1987	JAN 1988
WIND:														
Mean Speed (mph)	40	12.8	13.2	14.6	14.4	12.7	12.1	10.9	10.5	11.1	11.9	12.4	12.5	12.4
Prevailing Direction through 1963		N	N	SSE	SSE	SSE	SSE	SSE	SSE	SSE	SSE	S	S	SSE
Fastest Obs. 1 Min.														
-Direction (!!!)	7	36	35	12	31	33	33	32	27	21	36	26	32	21
-Speed (MPH)	7	43	36	39	46	46	38	31	35	52	37	46	33	52
-Year		1985	1988	1988	1985	1986	1987	1988	1988	1986	1985	1988	1987	SEP 1986
Peak Gust														
-Direction (!!!)	5	N	N	SW	NW	NW	N	NE	W	W	W	W	NW	W
-Speed (mph)	5	51	51	62	66	60	53	53	51	69	52	63	40	69
-Date		1985	1988	1985	1985	1986	1985	1987	1988	1987	1984	1988	1988	SEP 1987

(!!!) See Reference Notes on Page 68.
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PRECIPITATION (inches)

OKLAHOMA CITY, OKLAHOMA

YEAR	JAN	FEB	MAR	APR	MAY	JUNE	JULY	AUG	SEP	OCT	NOV	DEC	ANNUAL
1959	0.31	0.69	1.40	2.81	7.60	2.93	8.44	4.10	7.90	6.06	0.94	3.26	46.46
1960	1.47	2.12	0.85	3.01	6.79	5.68	7.69	2.89	1.62	4.66	0.24	2.48	39.50
1961	0.15	1.98	3.35	0.73	1.92	3.86	4.82	2.91	7.37	2.86	3.81	1.04	34.80
1962	1.45	1.02	0.80	2.16	2.64	7.84	1.71	2.26	3.08	2.43	1.34	0.76	27.49
1963	0.21	0.22	3.21	2.77	1.91	2.35	6.19	1.61	1.91	2.05	2.45	0.89	25.77
1964	0.83	2.17	1.30	2.06	5.21	0.77	2.01	4.91	2.96	0.84	5.46	0.62	29.14
1965	0.98	0.85	0.86	3.24	2.14	3.65	1.57	3.37	3.94	1.00	0.06	2.51	24.17
1966	1.05	2.39	1.30	3.68	0.88	2.63	2.38	6.77	2.82	0.37	0.84	0.45	25.56
1967	0.77	0.20	2.49	5.71	4.25	2.27	1.21	1.40	3.15	2.92	0.40	1.04	25.81
1968	2.19	1.02	2.84	3.03	8.40	2.39	1.41	3.75	2.64	2.40	4.11	1.33	35.51
1969	0.20	1.93	3.01	1.66	3.99	4.92	1.42	2.38	6.51	1.58	0.06	1.44	29.10
1970	0.32	0.29	2.09	5.33	6.53	2.45	1.30	0.80	9.64	3.29	1.03	0.26	33.33
1971	0.75	1.95	0.07	0.62	2.68	5.15	4.13	2.13	4.25	2.62	0.29	2.79	27.43
1972	0.21	0.43	1.13	3.10	4.03	1.36	3.22	1.82	2.04	7.17	2.28	0.84	27.63
1973	3.39	0.31	6.76	2.32	3.61	6.31	3.38	1.36	8.00	3.05	2.81	0.47	41.77
1974	0.10	2.68	3.12	4.66	5.01	3.36	0.48	4.42	6.24	5.57	2.34	1.47	39.45
1975	1.99	1.90	1.72	1.92	8.76	4.82	7.71	0.60	1.92	0.84	1.77	1.30	35.25
1976	T	0.33	3.09	2.94	4.36	0.88	1.38	1.46	1.53	1.78	0.12	0.19	18.06
1977	0.32	1.40	1.30	2.88	7.97	2.00	4.10	3.08	1.20	2.41	1.59	0.34	28.59
1978	1.26	3.23	1.32	1.65	10.12	4.04	3.75	0.25	0.96	1.02	2.88	0.70	31.18
1979	1.55	0.63	2.73	2.78	7.29	9.94	5.62	3.78	0.72	1.58	1.93	2.57	41.12
1980	1.69	1.29	1.38	2.16	9.00	2.52	0.42	0.60	2.21	0.99	0.51	1.58	24.35
1981	0.19	1.15	2.87	2.97	2.73	7.49	6.45	3.61	1.48	7.70	2.11	0.20	38.95
1982	3.68	0.98	1.63	1.92	12.07	4.06	2.11	1.13	2.86	1.03	2.78	1.94	36.19
1983	2.62	1.71	2.51	2.34	6.88	3.18	T	3.18	0.90	13.18	1.90	0.70	39.10
1984	0.35	1.16	4.70	1.79	1.62	3.48	0.30	2.35	1.01	6.64	2.05	8.14	33.59
1985	0.92	3.71	6.60	5.35	1.49	8.34	1.33	2.63	4.59	5.23	3.73	0.26	44.18
1986	0.00	0.68	1.75	4.42	8.21	3.11	0.38	3.29	9.54	8.00	4.63	1.16	45.17
1987	2.45	4.05	2.33	0.41	11.86	6.50	2.99	1.83	4.58	1.82	1.92	3.75	44.49
1988	1.24	0.41	7.85	3.19	1.07	3.59	1.92	1.60	5.19	2.04	2.45	1.39	31.94
Record Mean	1.26	1.27	2.25	3.18	5.21	4.00	2.71	2.65	3.23	2.98	1.93	1.49	32.15

See Reference Notes on Page 6B.
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AVERAGE TEMPERATURE (deg. F)

OKLAHOMA CITY, OKLAHOMA

YEAR	JAN	FEB	MAR	APR	MAY	JUNE	JULY	AUG	SEP	OCT	NOV	DEC	ANNUAL
1959	33.0	40.0	49.6	58.5	70.4	75.7	77.4	80.1	72.6	58.3	43.5	44.2	58.6
1960	36.3	34.5	39.7	61.7	66.5	77.3	78.2	79.3	74.5	64.4	51.4	37.4	58.4
1961	36.3	42.7	52.2	59.2	67.7	74.1	79.1	78.4	70.6	63.0	46.5	36.7	58.9
1962	32.3	45.1	48.1	58.3	74.3	75.3	82.2	82.5	71.4	64.7	49.4	40.6	60.4
1963	28.3	40.0	53.8	64.8	70.3	78.7	83.1	82.0	75.1	71.1	52.3	33.0	61.1
1964	40.1	38.5	47.1	64.1	70.0	77.4	85.3	80.8	72.3	60.3	49.9	37.9	60.3
*1965	38.8	39.5	40.7	65.5	71.3	78.0	84.6	80.6	73.9	62.8	56.3	48.8	61.8
1966	33.8	38.6	52.8	57.8	68.7	77.9	86.4	78.8	70.5	60.9	54.3	37.8	59.9
1967	41.8	41.7	56.4	65.4	66.9	77.4	79.7	79.3	70.9	62.6	49.4	39.8	60.9
1968	36.6	36.4	50.7	58.1	64.7	74.8	79.8	80.0	70.5	62.1	46.2	38.1	58.1
1969	38.8	42.3	42.0	60.4	67.7	75.0	83.9	80.2	73.3	57.6	48.7	40.4	59.2
1970	31.8	42.7	45.0	60.2	69.2	76.3	82.1	83.6	75.1	57.9	46.0	43.9	59.5
1971	36.9	39.1	49.1	60.4	67.3	78.6	80.7	77.2	73.3	63.4	49.0	42.2	59.8
1972	34.9	42.1	53.4	63.2	67.6	79.0	79.8	80.4	75.8	61.1	43.4	34.4	59.6
1973	33.3	39.8	52.5	56.0	66.8	75.2	79.8	79.7	70.6	64.3	53.1	39.3	59.2
1974	35.0	44.4	54.8	60.0	71.5	74.1	82.7	78.5	65.5	63.5	49.3	39.6	59.9
1975	40.3	36.5	46.1	58.7	67.4	75.1	78.0	80.1	68.3	63.4	50.7	41.8	58.8
1976	39.0	52.2	52.4	61.6	63.6	74.8	79.8	81.3	72.6	56.5	43.9	38.8	59.7
1977	29.2	45.9	54.1	62.5	70.0	79.6	83.0	80.7	78.0	62.7	50.9	40.0	61.4
1978	26.3	29.4	49.1	64.5	68.1	77.3	87.0	82.6	79.7	64.7	50.4	36.9	59.7
1979	25.4	31.5	51.2	58.1	65.8	75.2	81.0	80.0	73.1	65.7	46.5	43.3	58.1
1980	38.2	38.2	46.3	56.7	69.0	81.4	88.3	88.0	76.3	61.1	50.3	41.9	61.3
1981	37.7	43.9	51.9	65.6	65.7	78.4	84.2	78.8	74.1	60.1	50.3	39.1	60.8
1982	35.3	37.7	52.7	57.5	68.2	72.2	81.0	84.1	74.5	62.7	48.6	43.2	59.8
1983	38.6	42.6	48.8	54.0	64.6	73.4	81.6	84.0	74.9	62.7	50.4	25.8	58.5
1984	34.0	45.4	46.4	56.5	68.4	78.6	81.6	82.6	71.5	61.6	49.7	43.0	59.9
1985	30.6	37.2	53.0	62.7	70.0	76.0	80.9	81.3	73.1	61.2	46.1	35.1	58.9
1986	43.6	44.8	55.5	62.8	69.0	79.0	85.9	80.0	74.8	61.6	44.8	40.8	61.9
1987	35.1	45.9	50.3	61.8	72.6	77.1	80.1	82.2	72.4	60.0	50.5	40.6	60.7
1988	34.2	40.3	49.5	58.9	70.3	78.4	81.6	82.8	73.5	59.3	51.2	43.9	60.3
Record Mean	36.9	40.8	49.9	60.2	68.2	77.0	81.6	81.3	73.8	62.5	49.3	39.8	60.1
Max	46.9	51.5	61.3	71.3	78.3	87.2	92.4	92.4	84.7	73.6	59.9	49.6	70.8
Min	26.8	30.0	38.4	49.1	58.0	66.8	70.8	70.1	62.8	51.3	38.7	30.0	49.4

See Reference Notes on Page 6B.
Page 4B

HEATING DEGREE DAYS Base 65 deg. F

OKLAHOMA CITY, OKLAHOMA

SEASON	JULY	AUG	SEP	OCT	NOV	DEC	JAN	FEB	MAR	APR	MAY	JUNE	TOTAL
1959-60	0	0	24	206	640	644	880	877	777	134	73	0	4255
1960-61	0	0	2	96	407	852	885	618	395	247	49	0	3551
1961-62	0	0	40	106	547	868	1006	560	524	230	5	0	3886
1962-63	0	0	19	129	458	750	1131	692	372	101	50	0	3702
1963-64	0	0	7	23	381	988	761	762	547	116	27	6	3618
1964-65	0	0	24	150	454	831	808	706	745	71	2	0	3791
*1965-66	0	0	28	129	262	496	961	734	388	223	62	0	3283
1966-67	0	0	6	166	338	837	713	647	307	71	77	0	3162
1967-68	0	0	27	155	464	773	872	826	444	215	71	0	3847
1968-69	0	0	0	152	561	829	808	629	708	158	38	2	3885
1969-70	0	0	0	274	481	752	1022	620	615	187	31	12	3994
1970-71	0	0	18	254	559	651	866	718	492	163	36	0	3757
1971-72	0	0	59	88	475	702	923	660	365	144	46	0	3462
1972-73	0	0	23	225	640	940	975	701	380	283	55	0	4222
1973-74	0	0	37	99	362	787	922	573	330	168	8	0	3286
1974-75	0	0	56	88	463	784	763	792	583	235	29	0	3793
1975-76	0	0	64	126	430	713	801	367	406	128	100	0	3135
1976-77	0	0	19	306	629	805	1103	529	338	107	7	0	3843
1977-78	0	0	0	115	420	766	1192	990	493	90	64	0	4130
1978-79	0	0	2	89	437	866	1221	932	434	217	81	0	4279
1979-80	0	0	2	92	551	669	823	771	572	249	24	0	3753
1980-81	0	0	23	180	444	710	839	587	400	69	69	0	3321
1981-82	0	0	22	189	434	797	913	759	382	248	25	1	3782
1982-83	0	0	14	156	490	671	809	622	496	345	96	9	3708
1983-84	0	0	25	117	439	1207	955	561	572	263	45	0	4184
1984-85	0	0	75	162	462	676	1059	773	377	108	10	0	3702
1985-86	0	0	63	146	562	921	656	562	308	122	17	0	3357
1986-87	0	0	2	137	599	742	918	528	450	177	3	0	3556
1987-88	0	0	1	165	442	748	948	712	473	204	14	0	3707
1988-89	0	0	8	196	408	644						0	

See Reference Notes on Page 6B.
Page 5A

COOLING DEGREE DAYS Base 65 deg. F

OKLAHOMA CITY, OKLAHOMA

YEAR	JAN	FEB	MAR	APR	MAY	JUNE	JULY	AUG	SEP	OCT	NOV	DEC	TOTAL
1969	0	0	0	29	128	310	593	477	255	52	0	0	1844
1970	0	0	1	47	169	357	536	582	328	38	0	0	2058
1971	0	0	4	31	117	416	493	388	319	45	0	0	1810
1972	0	0	11	97	133	429	470	483	351	109	0	0	2085
1973	0	0	0	19	119	312	465	462	216	83	1	0	1687
1974	0	0	22	26	217	280	553	426	80	47	0	0	1651
1975	0	0	1	53	108	310	410	476	170	83	4	0	1615
1976	0	0	23	33	62	300	468	512	253	50	0	0	1702
1977	0	0	8	37	170	445	565	491	395	49	0	0	2163
1978	0	0	8	80	165	378	690	553	450	87	2	0	2418
1979	0	0	10	18	112	314	505	471	252	121	2	0	1805
1980	0	0	0	7	155	498	729	721	366	65	1	0	2554
1981	0	0	4	94	98	409	603	435	304	47	0	2	1994
1982	0	0	9	28	130	234	503	598	305	90	3	1	1901
1983	0	0	0	20	91	266	523	599	329	54	8	0	1890
1984	0	0	0	16	159	414	521	551	279	64	5	0	2009
1985	0	0	12	43	172	336	501	512	313	38	0	0	1927
1986	0	0	21	63	147	425	653	473	301	40	0	0	2125
1987	0	0	0	88	242	371	475	543	230	18	12	0	1979
1988	0	0	1	29	186	410	525	558	270	25	1	0	2005

See Reference Notes on Page 6B.
Page 5B

SNOWFALL (inches)

OKLAHOMA CITY, OKLAHOMA

SEASON	JULY	AUG	SEP	OCT	NOV	DEC	JAN	FEB	MAR	APR	MAY	JUNE	TOTAL
1959-60	0.0	0.0	0.0	0.0	0.3	T	8.2	5.4	3.2	0.0	0.0	0.0	17.1
1960-61	0.0	0.0	0.0	0.0	0.0	8.2	1.9	7.2	0.7	T	0.0	0.0	18.0
1961-62	0.0	0.0	0.0	0.0	0.0	0.8	8.5	0.8	0.0	0.0	0.0	0.0	10.1
1962-63	0.0	0.0	0.0	0.0	T	1.2	0.5	1.6	0.0	0.0	0.0	0.0	3.3
1963-64	0.0	0.0	0.0	0.0	0.0	2.5	T	0.6	1.5	0.0	0.0	0.0	4.6
1964-65	0.0	0.0	0.0	0.0	0.0	T	2.0	0.9	0.1	0.0	0.0	0.0	3.0
1965-66	0.0	0.0	0.0	0.0	0.0	T	5.8	2.3	0.0	0.0	0.0	0.0	8.1
1966-67	0.0	0.0	0.0	0.0	T	0.6	T	0.1	0.4	0.0	0.0	0.0	1.1
1967-68	0.0	0.0	0.0	T	1.2	1.2	0.4	7.7	13.9	0.0	0.0	0.0	24.4
1968-69	0.0	0.0	0.0	0.0	1.4	1.3	T	1.3	8.2	0.0	0.0	0.0	12.2
1969-70	0.0	0.0	0.0	0.0	0.0	4.2	0.4	T	2.3	T	0.0	0.0	6.9
1970-71	0.0	0.0	0.0	0.0	T	T	T	5.1	0.7	0.0	0.0	0.0	5.8
1971-72	0.0	0.0	0.0	0.0	0.5	5.2	0.8	4.9	0.0	0.0	0.0	0.0	11.4
1972-73	0.0	0.0	0.0	0.0	7.5	1.4	8.3	0.2	T	T	0.0	0.0	17.4
1973-74	0.0	0.0	0.0	0.0	0.0	0.6	0.7	1.0	0.5	0.0	0.0	0.0	2.8
1974-75	0.0	0.0	0.0	0.0	1.0	2.0	0.6	0.9	0.1	0.0	0.0	0.0	4.6
1975-76	0.0	0.0	0.0	0.0	0.7	3.9	T	0.3	0.0	0.0	0.0	0.0	4.9
1976-77	0.0	0.0	0.0	0.0	0.3	T	2.8	0.4	0.0	0.0	0.0	0.0	3.5
1977-78	0.0	0.0	0.0	0.0	T	0.0	8.4	12.0	T	0.0	0.0	0.0	20.4
1978-79	0.0	0.0	0.0	0.0	0.0	3.3	4.0	6.1	0.0	0.0	0.0	0.0	13.4
1979-80	0.0	0.0	0.0	0.0	T	T	T	1.8	T	0.0	0.0	0.0	1.8
1980-81	0.0	0.0	0.0	0.0	4.0	0.0	T	T	0.0	0.0	0.0	0.0	4.0
1981-82	0.0	0.0	0.0	0.0	0.0	T	1.0	3.9	2.5	0.0	0.0	0.0	7.4
1982-83	0.0	0.0	0.0	0.0	T	T	5.1	4.3	T	0.0	0.0	0.0	9.4
1983-84	0.0	0.0	0.0	0.0	T	1.9	5.6	2.0	T	0.0	0.0	0.0	9.5
1984-85	0.0	0.0	0.0	0.0	T	6.1	1.5	2.3	0.0	0.0	0.0	0.0	9.9
1985-86	0.0	0.0	0.0	0.0	T	2.9	0.0	10.9	0.0	0.0	0.0	0.0	13.8
1986-87	0.0	0.0	0.0	0.0	0.0	T	10.0	1.0	T	0.0	0.0	0.0	11.0
1987-88	0.0	0.0	0.0	0.0	2.0	8.3	12.1	0.2	0.9	0.0	0.0	0.0	23.5
1988-89	0.0	0.0	0.0	0.0	0.6	2.0							
Record Mean	0.0	0.0	0.0	T	0.5	1.7	3.1	2.6	1.4	T	0.0	0.0	9.4

See Reference Notes on Page 6B.
Page 6A

REFERENCE NOTES

OKLAHOMA CITY, OKLAHOMA

GENERAL

T - TRACE AMOUNT.
BLANK ENTRIES DENOTE MISSING/UNREPORTED DATA.
INDICATES A STATION OR INSTRUMENT RELOCATION.
SEE STATION LOCATION TABLE ON PAGE 8.

SPECIFIC

PAGE 2
PM - INCLUDES LAST DAY OF PREVIOUS MONTH

PAGE 3

(*) - LENGTH OF RECORD IN YEARS, ALTHOUGH INDIVIDUAL MONTHS MAY BE MISSING.

* LESS THAN .05
NORMALS - BASED ON THE 1951-1980 RECORD PERIOD.
EXTREMES - DATES ARE THE MOST RECENT OCCURRENCE.
WIND DIR. - NUMERALS SHOW TENS OF DEGREES CLOCKWISE FROM TRUE NORTH. "00" INDICATES CALM.
RESULTANT DIRECTIONS ARE GIVEN TO WHOLE DEGREES.

PAGE 4B

MAX AND MIN ARE LONG TERM MEAN DAILY MAXIMUM AND MEAN DAILY MINIMUM TEMPERATURES.

EXCEPTIONS

PAGE 3

1. FASTEST MILE WIND IS THROUGH OCTOBER 1981.
PAGES 4A, 4B, 6A
RECORD MEANS ARE THROUGH THE CURRENT YEAR, BEGINNING IN 1891 FOR TEMPERATURE
1891 FOR PRECIPITATION
1940 FOR SNOWFALL

OKLAHOMA CITY, OKLAHOMA

Oklahoma City is located along the North Canadian River, a frequently nearly-dry stream, at the geographic center of the state. It is not quite 1,000 miles south of the Canadian Border and a little less than 500 miles north of the Gulf of Mexico. The surrounding country is gently rolling with the nearest hills or low mountains, the Arbuckles, 80 miles south. The elevation ranges around 1,250 feet above sea level.

Although some influence is exerted at times by warm, moist air currents from the Gulf of Mexico, the climate of Oklahoma City falls mainly under continental controls characteristic of the Great Plains Region. The continental effect produces pronounced daily and seasonal temperature changes and considerable variation in seasonal and annual precipitation. Summers are long and usually hot. Winters are comparatively mild and short.

During the year, temperatures of 100 degrees or more occur on an average of 10 days, but have occurred on as many as 50 days or more. While summers are usually hot, the discomforting effect of extreme heat is considerably mitigated by low humidity and the prevalence of a moderate southerly breeze. Approximately one winter in three has temperatures of zero or lower.

The length of the growing season varies from 180 to 251 days. Average date of last freeze is early April and average date of first freeze is early November. Freezes have occurred in early October.

During an average year, skies are clear approximately 40 percent of the time, partly cloudy 25 percent, and cloudy 35 percent of the time. The city is almost smoke-free as a result of favorable atmospheric conditions and the almost exclusive use of natural gas for heating. Flying conditions are generally very good with flight by visual flight rules possible about 96 percent of the time.

Summer rainfall comes mainly from showers and thunderstorms. Winter precipitation is generally associated with frontal passages. Measurable precipitation has occurred on as many as 122 days and as few as 55 days during the year. The seasonal distribution of precipitation is normally 12 percent in winter, 34 percent in spring, 30 percent in summer, and 24 percent in fall. The period with the least number of days with precipitation is November through January, and the month with the most rainy days is May. Thunderstorms occur most often in late spring and early summer. Large hail and/or destructive winds on occasion accompany these thunderstorms.

Snowfall averages less than 10 inches per year and seldom remains on the ground very long. Occasional brief periods of freezing rain and sleet storms occur.

Heavy fogs are infrequent. Prevailing winds are southerly except in January and February when northerly breezes predominate.

STATION LOCATION

Location	Occupied from	Occupied to	Airline distance and direction from previous location	Latitude North	Longitude West	Elevation above								Type M = AMOS T = AUTOS	Remarks
						Sea level	Ground								
							Ground at temperature site	Wind instruments	Extreme thermometers	Psychrometer	Sunshine switch	Tipping bucket rain gage	Weighting rain gage		
CITY															
Overholser Opera House SE Corner, Robinson and Grand	11/01/90	6/30/02		35° 28'	97° 30'	1196	53	54	54		45		45		
Culbertson Building SE Corner, Broadway and Grand	7/01/02	1/16/06	400 feet E	35° 28'	97° 30'	1196	86	79	79		71		71		
WB Observatory 1923 Classen Boulevard	1/17/06	3/01/55	1.9 mile NW	35° 29'	97° 32'	1254		10				5			
AIRPORT															
Administration Building Will Rogers Field 7 miles SW of PO	4/02/32	5/20/54		35° 24'	97° 36'	1280	70	33	31		4	4	4		
Page Hanger #3 Will Rogers Field	5/21/54	10/21/65	3/16 mile N	35° 24'	97° 36'	1280	55	7	6		4	5	4	a5	
Weather Bureau Building Will Rogers World AP	10/21/65	Present	200 feet NE	35° 24'	97° 36'	1285	20	7 c5	6 c5	222 124	4	5	4	b5 d5 e5	NA
†Nat. Weather Service Bldg. 1970.															
							</								

Offices consolidated at
Airport 4/1/54.

a - Telepsychrometer(6'') 5/21/54-
12/31/60. Hygro. comm. 12/31/60.
b - 1 mile south of previous site.
c - Effective December 1976.
d - Commissioned 10/20/65.
e - Type change 5/6/83.
f - Type change 12/6/85.
WSFO moved to 1200 Westheimer
Drive, Norman, OK 1/20/87.
WSO established at AP 3/11/87.
f - Minor adjustment mid-1987.

SUBSCRIPTION:
Price and ordering information available through: National Climatic Data Center, Federal Building, Asheville, North Carolina 28801

USCOMB-NOAA-ASHEVILLE, N.C. - 470

U.S. DEPARTMENT OF COMMERCE
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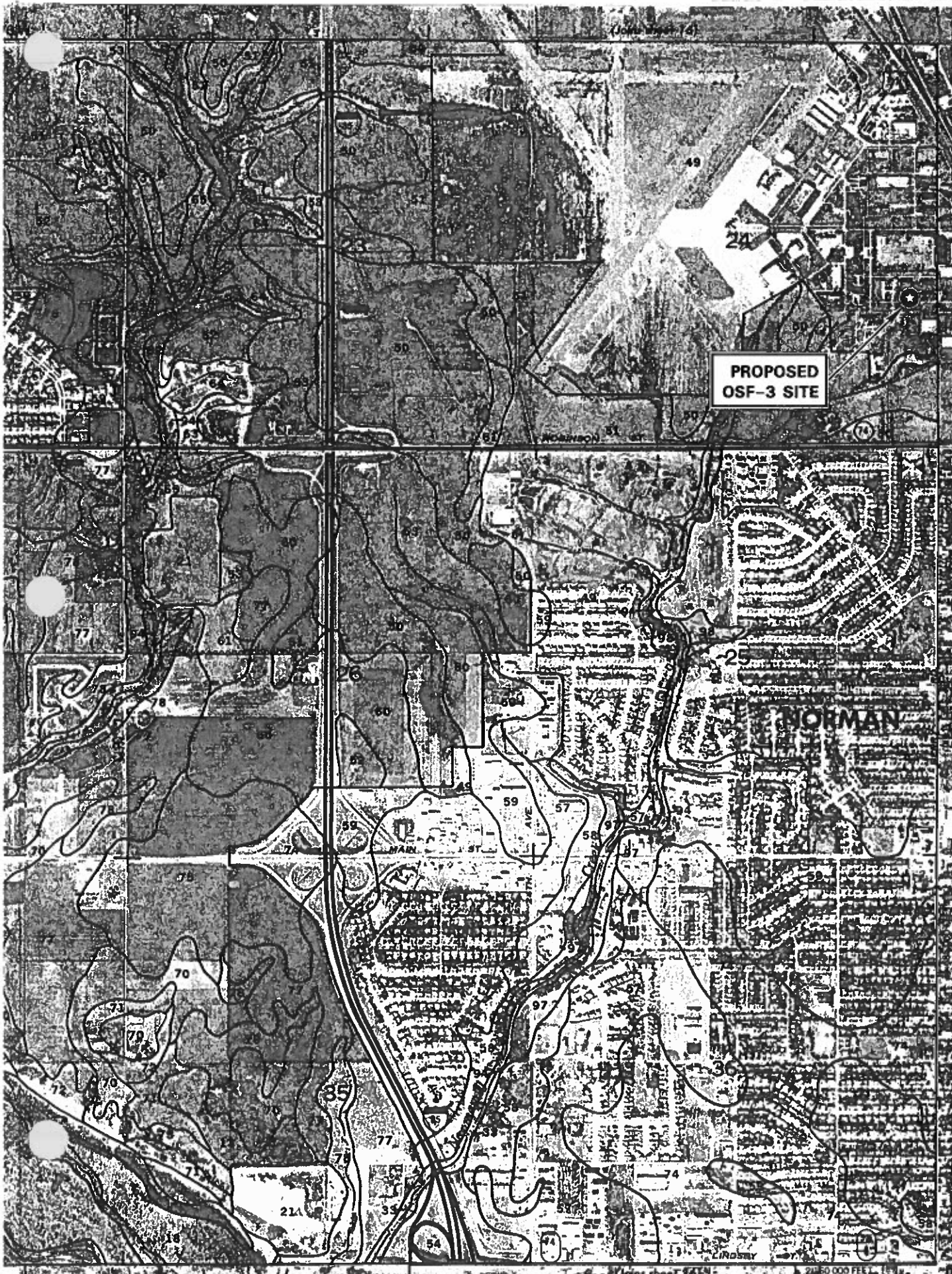


FIRST CLASS

49 DOOLIN-URBAY LAVO-PAWUSKA COMPLEX, 0 TO 3 PERCENT SLOPES
DOOLIN PART MAKES UP ABOUT 40 PERCENT OF COMPLEX

ESTIMATED SOIL PROPERTIES															
DEPTH (IN.)	USDA TEXTURE	UNIFIED	AASHTO	PERCENT OF MATERIAL LESS NO. 20 SIEVE	PERCENT OF MATERIAL LESS NO. 40 SIEVE	PERCENT OF MATERIAL LESS NO. 60 SIEVE	PERCENT OF MATERIAL LESS NO. 100 SIEVE	PERCENT OF MATERIAL LESS NO. 200 SIEVE	LIQUID LIMIT	PLASTICITY INDEX	PERCENT PASSING NO. 20 SIEVE	PERCENT PASSING NO. 40 SIEVE	PERCENT PASSING NO. 60 SIEVE	PERCENT PASSING NO. 100 SIEVE	PERCENT PASSING NO. 200 SIEVE
0-12	SIL	CL, CL-ML	A-4, A-6	100	100	100	100	100	24-37	4-14	100	100	100	100	100
12-30	SIL	CL, CL	A-4, A-6	100	100	100	100	100	24-37	4-14	100	100	100	100	100
30-42	SIL	CL, CL	A-4, A-6	100	100	100	100	100	24-37	4-14	100	100	100	100	100
42-60	SIL	CL, CL	A-4, A-6	100	100	100	100	100	24-37	4-14	100	100	100	100	100
60-72	SIL	CL, CL	A-4, A-6	100	100	100	100	100	24-37	4-14	100	100	100	100	100
72-84	SIL	CL, CL	A-4, A-6	100	100	100	100	100	24-37	4-14	100	100	100	100	100
84-96	SIL	CL, CL	A-4, A-6	100	100	100	100	100	24-37	4-14	100	100	100	100	100
96-108	SIL	CL, CL	A-4, A-6	100	100	100	100	100	24-37	4-14	100	100	100	100	100
108-120	SIL	CL, CL	A-4, A-6	100	100	100	100	100	24-37	4-14	100	100	100	100	100
120-132	SIL	CL, CL	A-4, A-6	100	100	100	100	100	24-37	4-14	100	100	100	100	100
132-144	SIL	CL, CL	A-4, A-6	100	100	100	100	100	24-37	4-14	100	100	100	100	100
144-156	SIL	CL, CL	A-4, A-6	100	100	100	100	100	24-37	4-14	100	100	100	100	100
156-168	SIL	CL, CL	A-4, A-6	100	100	100	100	100	24-37	4-14	100	100	100	100	100
168-180	SIL	CL, CL	A-4, A-6	100	100	100	100	100	24-37	4-14	100	100	100	100	100
180-192	SIL	CL, CL	A-4, A-6	100	100	100	100	100	24-37	4-14	100	100	100	100	100
192-204	SIL	CL, CL	A-4, A-6	100	100	100	100	100	24-37	4-14	100	100	100	100	100
204-216	SIL	CL, CL	A-4, A-6	100	100	100	100	100	24-37	4-14	100	100	100	100	100
216-228	SIL	CL, CL	A-4, A-6	100	100	100	100	100	24-37	4-14	100	100	100	100	100
228-240	SIL	CL, CL	A-4, A-6	100	100	100	100	100	24-37	4-14	100	100	100	100	100
240-252	SIL	CL, CL	A-4, A-6	100	100	100	100	100	24-37	4-14	100	100	100	100	100
252-264	SIL	CL, CL	A-4, A-6	100	100	100	100	100	24-37	4-14	100	100	100	100	100
264-276	SIL	CL, CL	A-4, A-6	100	100	100	100	100	24-37	4-14	100	100	100	100	100
276-288	SIL	CL, CL	A-4, A-6	100	100	100	100	100	24-37	4-14	100	100	100	100	100
288-300	SIL	CL, CL	A-4, A-6	100	100	100	100	100	24-37	4-14	100	100	100		

B-10



MEMORANDUM

SRI GEOSCIENCE AND ENGINEERING CENTER

TO: Patti Burns
FROM: Juris Petriceks 
SUBJECT: Multiple RF Sources
DATE: August 3, 1994
LOCATION: G-229
cc: Dave Elsner

General

At some locations members of the general public will be exposed to radio-frequency radiation (RFR) from two (or more) radars and from other RF sources (television and radio transmitters, cellular telephones, garage-door openers,...). Exposure to RFR from all sources must be considered to ensure that no health hazard occurs.

The ANSI/IEEE (1992) standard is accepted in the U.S. for safe exposure levels to RFR. The maximum permissible exposure levels as a function of the frequency of the RFR source is given in Tables 1 and 2 of the April 27, 1992 document "IEEE Standard for Safety Levels with Respect to Human Exposure to Radio Frequency Electromagnetic Field, 3 kHz to 300 GHz." For the WSR-88D frequency band (2.7 to 3.0 GHz) the maximum permissible exposure (MPE) level for uncontrolled environments (accessible to the public) is 1.8 to 2.0 mW/cm². For controlled environments (not accessible to the public) the MPE level is five times higher (9.0 to 10.0 mW/cm²). For television and FM radio transmitters in the 100 to 300 MHz frequency band the MPE level for uncontrolled environments is 0.2 mW/cm² (for controlled environments—1.0 mW/cm²). These MPE values are derived by averaging the RFR power density over a time interval of thirty minutes for uncontrolled environments (six minutes for controlled environments).

For exposure to pulsed RFR in the 0.1 to 300,000 MHz frequency range, the ANSI/IEEE (1992) standard provides two ways to determine the peak (temporal) value of MPE. The strictest of these methods yields a value of 2.65×10^8 mW/cm². For the WSR-88D transmitted RF power, pulse lengths, and pulse repetition frequencies, the average power density MPE is the controlling value. For television and radio transmitters the average and peak power density levels are the same, and again the average power density MPE is the controlling value.

When a number of RFR sources contribute to the total exposure, it becomes necessary to weigh each contribution relative to the MPE. The fraction of the MPE contributed by each source should be determined and the sum of all such fractions should not exceed unity. For example if the RFR exposure at any particular location to each of two WSR-88Ds were 0.18 mW/cm², each would contribute 0.1 of the MPE (0.18 mW/cm² divided by 1.8 mW/cm²) for a total of 0.2 of the MPE. Adding a third WSR-88D that resulted in the same RFR exposure of 0.18 mW/cm² would increase the total to 0.3 of the MPE. However, if the third source were a television transmitter that resulted in RFR exposure of 0.18 mW/cm², that would add 0.9 of the MPE (0.18 mW/cm² divided by 0.2 mW/cm²) and would increase the total to 1.1 of the MPE - an unacceptable situation because the total exceeds the MPE.

Unlike ionizing radiation, such as X rays, non-ionizing radiation, such as RFR, does not have a cumulative effect of day-to-day or year-to-year exposure.

One WSR-88D (KOUN) on a 20-m tower has been located at Max Westheimer Airpark since 1989. Another WSR-88D (OSF-3) on a 30-m tower is planned to be located at Max Westheimer Airpark.

The following table shows the RFR power density levels from the two radars at two public buildings and adjacent to the radars. These levels are obtained from the "Supplemental Environmental Assessment (SEA) of the Effects of Electromagnetic Radiation from the WSR-88D Radar".

RFR POWER DENSITY (mW/cm ²)		
LOCATION	SOURCE	
	KOUN	OSF-3
YMCA	0.00002	0.00003
RED CROSS	0.00005	0.00001
KOUN	0.0006	0.00002
OSF-3	0.00003	0.0003

The highest RFR power density is encountered adjacent to the KOUN WSR-88D; the KOUN WSR-88D contributes 1/3,000 of the MPE and the OSF-3 WSR-88D contributes 1/90,000 of the MPE for a total of 1/2,900 of the MPE. The highest RFR power density at a building frequented by the general public is at the Red Cross Building; the OSF-3 WSR-88D contributes 1/180,000 of the MPE and the KOUN WSR-88D contributes 1/36,000 of the MPE for a total of 1/30,000 of the MPE. The highest RFR power density levels are well below the MPE.

There are no television or radio transmitter towers close enough to the WSR-88Ds that would contribute RFR power density levels approaching the MPE. Devices such as garage-door openers and cellular telephones are capable of producing relatively large near-in RFR power density levels even though their powers are quite modest. The relevant point is that the small fractions of MPE contributed by the two WSR-88D systems will add only insignificantly to the near-in power density levels produced by such devices.

MEMORANDUM

SRI GEOSCIENCE AND ENGINEERING CENTER

TO: Juris Petriceks
FROM: Bill Edson *Bill Edson*
SUBJECT: OSF-3 at Norman, OK

DATE: May 8, 1994

LOCATION: G-238

cc: Patti Burns
Dave Elsner

Introduction

A new Operational Support Facility (OSF) WSR-88D--OSF-3--is to be built at Norman, Oklahoma, and the existing OSF WSR-88D, (here designated KOUN), is to be transferred to the custody of the National Severe Storms Laboratory (NSSL). For a variety of reasons it is desirable to locate the new radar close to the OSF and, hence, to the existing KOUN. The following paragraphs discuss the problem of choosing a location and tower height for OSF-3 that optimize its operation without significantly deteriorating that of the KOUN, and selecting a frequency that avoids interference with existing radars. The choice of frequency is complicated by the presence of a large number of S-band radars in the surrounding area--including KOUN, the WSR-88D at Twin Lakes, and the Norman Doppler Radar (NDR), which is expected to cease operations. The choice of location is complicated by potential beam blocking caused by two water tanks, the MLOS tower and the other radars.

General

The blockage of the KOUN beam will be minimized if OSF-3 is located in the direction of one of the water tanks. Therefore, the three candidate sites, designated A, B, and C, are located along these axes. This constraint does not substantially affect the blockage of the OSF-3 beam. Both water tanks are similar, having a cylindrical section with conical top and bottom sections. Both are supported on multi-leg towers that are relatively open. The blocking area of each tower is reasonably well approximated by a sphere, 50 ft in diameter. The MLOS tower is quite open and about 3 ft wide, with an effective width of about 0.75 ft. Many calculations involve the diameter of the WSR-88D antenna, which is 28 ft.

Tower Height

Tower heights of 5, 10, 15, 20, 25, and 30 meters are available to WSR-88D systems. Of these, 20 m is undesirable in the present case because that value is used by KOUN. Use of that value would lead to excessive mutual Electromagnetic Interference (EMI). Tower heights of 15 and 25 m are undesirable for the same reason, but to a lesser extent. A tower height of 5 m is undesirable on the basis of coverage and potential health hazard considerations. Therefore, only tower heights of 10 and 30 meters are analyzed here. A 10-m tower places the antenna center 48 ft above the base; a 30-m tower places it at 114 ft.

Locations

The locations of candidate sites A, B, and C are shown in Figure 1. For reference the coordinates of KOUN are 35° 14' 10" N and 97° 27' 47" W (NAD 1927).

Analysis

The relative merits of the proposed locations and tower heights are best analyzed and compared by means of a suitable table. Accordingly the key parameters of the various locations are gathered in Table I. The first line gives the elevation of the ground at the designated location. The second line gives the elevation of the top of the structure; e.g., the upper edge of the WSR-88D antennas, the tops of the water tanks, and the top of the MLOS tower. The third line gives the elevations of the centers of the antennas or water tanks. Lines four to seven give relevant distances, including several redundant entries. These values of distance and elevation are derived from the applicable topographic map. Line eight gives, for each location, the elevation of the center of the lowest (+0.5°) beam from KOUN. This figure is derived by noting that the beam rises about one foot in each 120 ft traveled.

Beam Blockage

With these data it is possible to estimate the degree of beam blockage and EMI experienced by both radars for each of the candidate locations. It is convenient to start with KOUN because the various locations are referred to it. Estimates of blockage are based on Figures 4 and 5 of my memorandum "Beam Blockage of the WSR-88D Radar" dated November 4, 1993.

The blockage caused to KOUN by the MLOS tower and the two water tanks is estimated first, and will serve as a basis for comparing the increase caused by the OSF-3. Values derived from curves of the cited Figures 4 and 5 are given in line nine of Table I. In the far field of the WSR-88D beam the first nulls are separated by 3.0 deg. Thus, an obstruction that subtends an angle of θ deg at the source produces some blockage over a total angle of $\theta + 3 + 3$ deg. The near-field column is taken as a cylinder 28 ft in diameter. Twice this value, supplemented by the width of the obstacle and related to the distance, yields the azimuth blockage of nearby objects. Results of these calculations are given in line eleven. The elevation angle at which the greatest beam blockage occurs is found by taking arc tan of the applicable elevation difference divided by the distance. Two values are given when the result is about midway between two standard elevation scan angles. Also, the estimate of blockage is somewhat reduced for these cases. For Site A the KOUN beam blockage in the direction of the NE water tank will be a composite of that produced by the OSF-3 antenna and the tank, while other blockages are unchanged. For Sites B and C the KOUN blockage will be a composite in the direction of the SE tank, while other blockages are unchanged. If the 30-m tower is used these additional blockages will be time variable because of the rotation of the OSF-3 antenna, and will be quite substantial. If the 10-m tower is used, they will be negligible compared to the existing blockage, and are designated zero in the table.

Of equal importance, is potential blockage of the OSF-3 beam which is estimated in the same way. If the 30-m tower is used at any of the three sites, the OSF-3 beam will pass well above the top of the KOUN antenna and will not suffer blockage from it. If the 10-m tower is used, the KOUN will produce a substantial and time variable blockage. The MLOS and two water towers are also considered. The most unfavorable elevation angle and resulting blockage appear as dual entries in lines twelve through fifteen.

Intercepted Power

When two similar radars must operate in close proximity there is always a risk that the power delivered by one to the other will be sufficient to damage front-end components. My memorandum "Near Field Power Densities of the WSR-88D Radar" includes curves that provide an estimate of intercepted power levels. In each case, the capture area of one radar is multiplied by the power density delivered to it by the other. By reciprocity, the power transfer in the other direction is the same. With the 10-m tower the OSF-3 antenna is always lower than KOUN, but will occasionally "look up" directly toward it. The upward directed KOUN beam passes above OSF-3, and the difference of elevations yields the beam offset, which with the horizontal distance, yields the power density. With the 30-m tower the situation is reversed but the procedure is the same.

The physical area of a 28-ft circular aperture is $(14 \times 30.48)^2 \pi \text{ cm}^2 = 5.78 \times 10^5 \text{ cm}^2$. This value is reduced by a factor near two to 3×10^5 to account for imperfect antenna efficiency. The curves of Figure 3 of the cited memorandum yield power densities at various locations, the upper and lower edges of the receiving antenna being used here. The higher density at the upper edge exists only over a small fraction of the total area. Therefore, a value considerably smaller than the simple average is used as the weighted average.

The power levels listed in line eighteen range from 300 to 8000 watts, all sufficiently high to damage front-end components. A waveguide filter with 70 dB discrimination would reduce these values to the range 0.03 to 0.8 mW--values that are deemed safe. A filter of this type is needed to avoid interference as discussed in the section "Choice of Frequency".

Health Hazards

For the 10-m tower the axis of the +0.5 deg beam starts at an elevation of 48 ft AGL and rises one foot in each 120 ft traveled. The American Red Cross building is about 1300 ft south of Site A, and the YMCA building is about 500 ft to the northwest; both are treated as two-story structures. There are many other similar buildings in the vicinity. In the unlikely event that the OSF-3 antenna should stop rotating and radiate full power at 0.5 deg elevation the average power density on the beam axis would be $0.0021 \times 4250 = 8.9 \text{ mW/cm}^2$ for distances out to a few hundred feet. The average power density decreases to 1.8 mW/cm^2 (the ANSI/IEEE (1992) value for uncontrolled environments) at about 1300 ft. For distances greater than 18 ft below the beam axis the average power density never exceeds 0.4 mW/cm^2 . The WSR-88D system contains interlocks to prevent radiation if the antenna ceases to rotate, and these interlocks are deliberately disabled only rarely when the antenna is pointed in a safe direction to permit system testing. Even in the event of erroneous operation with a fixed beam there would be no health hazard at the YMCA or Red Cross buildings, or to any individuals less than 30 ft above ground level. The same statements apply to the case of the 30-m tower, except that the above-ground level of safety rises to 96 ft. Similar statements apply to sites B and C.

Choice of Frequency

The frequency of the KOUN system is 2995 MHz, the NDR uses 2850 MHz, and 2910 MHz is used by the radar at Twin Lakes, which is about 12 (statute) miles to the northeast. These frequencies have been confirmed by reference to a computer printout prefaced by the Electromagnetic Compatibility Analysis Center (ECAC) dated April 6, 1994. The NDR system is rarely used and is scheduled to be removed by the time that the OSF-3 becomes functional.

The cited ECAC printout shows that the frequency band from 2700 to 2900 MHz is heavily occupied by radar emissions. In the geographical area of interest there are 50 assignments between 2700 and 2800 MHz, and an additional 64 are in the band 2800 to 2900 MHz. In contrast, there are only 7 assignments between 2900 and 3000 MHz. Of the latter, two are occupied by KOUN and Twin Lakes. The other five are assigned to the U.S. Army at Fort Sill, OK, located at $34^\circ 40' 24'' \text{ N}$; $98^\circ 23' 52'' \text{ W}$, which is more than 60 miles west of Norman. These frequencies range from 2921 to 2929 MHz and are radiated at 200 kW.

At first glance it appears logical to choose for OSF-3 the frequency 2850 MHz which is expected to be released by the NDR. However, the ECAC printout lists an ASR-8 at 2835 and 2863 MHz, and an ASR-9 at 2840 MHz, all at Will Rogers Airport about 13 miles to the northwest. At Tinker AFB there are AN/TPN 19 and 24 systems that use 2875 MHz, also about 13 miles distant. At Norman there are AN/TPN 19 and 24 systems that use 2825 MHz. Page 10, Figure 5 of the report, "NEXRAD In-Depth Site Survey for the Operational Support Facility (OSF) NORMAN, OKLAHOMA," dated April 1987, confirms the presence of many strong signals at this site, especially in the range 2825 - 2840 MHz.

To estimate the interference that the ASR-9, 13 miles away at 2840 MHz might cause if 2850 MHz were chosen for OSF-3 we refer to the SRI report "NEXRAD EMI Study", April 1984. Table 2, page 26 of that report, shows that 193 dB is the Total Required Loss to prevent radiation from an ASR-9 radar interfering with a NEXRAD receiver. Page 16 gives the value of the free-space path loss for 13 miles as $105.5 + 20$

$\log_{10} R = 128$ dB. The difference $193 - 128 = 65$ dB must be provided by Frequency-Dependent Rejection. Figure 4, page 18, which is applicable to the $1.05\text{-}\mu\text{sec}$ pulse of the ASR-9, shows that a 10 MHz frequency difference provides only 40 dB rejection, which implies 25 dB interference. Intervening foliage and earth curvature are unlikely to provide enough attenuation to overcome this level of interference.

The AN/TPN 19 and 24 systems are closer and use magnetrons which produce wider spectra than the ASR-9 at Will Rogers Airport. They too are regarded as potentially serious sources of EMI.

The frequencies chosen for Twin Lakes and KOUN reflect the fact that the frequency band above 2900 MHz is much less densely occupied than the band below 2900 MHz. For the same reason it is recommended that OSF-3 use a frequency roughly midway between 2910 and 2995 MHz. The preferred value is 2950 MHz, with 2945 and 2955 as acceptable alternatives. The choice of the frequency is dominated by the frequency difference required to prevent EMI. Therefore, a review of that topic is appropriate. The cited SRI report also considers the case of interference between two NEXRAD (WSR-88D) systems. Assuming that both use waveguide bandpass filters of the type commonly used in ASR-8 and ASR-9 systems, a frequency difference of 40 MHz is sufficient if the separation distance exceeds 700 ft. (See Fig 6, p 22 of the report.)

The 2955 value provides a 40-MHz offset from the close-by KOUN. The 2945 value provides a 16-MHz offset from the distant and less powerful Army radar at Fort Sill (earth curvature is relied upon to add to the path loss for this case).

The Intermediate Frequency

The intermediate frequency (IF) of the WSR-88D is 57.55 MHz, and the local oscillator frequency is below the receive frequency. Based on these facts, there is a legitimate concern that interference might result from a second radar operating nearby at the image frequency 57.55 MHz below the local oscillator and hence 115.1 MHz below the assigned frequency. To prevent such a response it is necessary to provide strong frequency discrimination in front of the mixer. The recommended frequency allocation avoids this difference.

Summary and Conclusions

1. Site A, 900 ft northeast of KOUN on ground at an elevation of 1177 ft near Westheimer Drive is favored.
2. A frequency of 2950 MHz is recommended with 2945 and 2955 as acceptable alternatives.
3. A tower height of 30 m is preferred because it minimizes blockage to the OSF-3 and provides the lowest possible radio frequency radiation power densities at ground level and nearby buildings. This choice somewhat increases the degree of beam blockage of KOUN in the direction of the water tower and increases construction costs.
4. A tower height of 10 m is acceptable. It reduces cost and the extent of KOUN beam blockage in the direction of the water tower at the cost of creating blockage to the OSF-3 in the KOUN direction and increasing the radio frequency and radiation power densities at ground level and nearby buildings, although still 900 times lower than the ANSI/IEEE (1992) standard for average power density.
5. The MLOS tower will continue to produce a small amount of blockage to the KOUN beam and will provide a similar blockage to the new OSF-3 beam for either tower height.
6. Damage to front-end components is a serious risk, and EMI is a potential problem. A highly selective high power filter is needed for both OSF-3 and KOUN to solve this problem.
7. Sites B and C are less convenient for the staff, and may involve increased construction and utility costs, but are functionally acceptable.

TABLE 1
Parameters for Choice of OSF-3 Location

				Site A				Site B				Site C			
				KOUN	MLOS Tower	NE Water Tower	SE Water Tower	10-m Tower	30-m Tower	10-m Tower	30-m Tower	10-m Tower	30-m Tower	10-m Tower	30-m Tower
1	Elevation/Base Above MSL	feet	1170	1177	1184	1193	1179	1179	1179	1178	1178	1182	1182	1182	1182
2	Elevation/Top of structure	feet	1166	1328	1337	1321	1241	1241	1307	1242	1306	1246	1310	1310	1310
3	Elevation/Center of Antenna or Water Tower	feet	1252	N/A	1312	1296	1227	1227	1293	1226	1292	1230	1296	1296	1296
4	Distance from KOUN	feet	0	700	2300	2900	900	900	900	1100	1100	1450	1450	1450	1450
5	Distance from Site A	feet	900	300	1400	2700	0	0	0	1200	1200	1400	1400	1400	1400
6	Distance from Site B	feet	1100	900	2150	1800	1200	1200	1200	0	0	350	350	350	350
7	Distance from Site C	feet	1450	1150	2200	1450	1400	1400	1400	350	350	0	0	0	0
8	Elevation of KOUN Beam center at Site	feet		1258	1272	1277	1260	1260	1260	1262	1262	1264	1264	1264	1264
9	Estimated Beam Blockage of KOUN	dB		0.3 to 0.4	7	4	0	0	18	0	10	0	5	5	5
10	Elevation of Beam Subject to Greatest Blockage	deg		0 to 6	1.5	0.5	none	none	2.5	none	1.5, 2.5	none	1.5	1.5	1.5

11	Total Azimuth Angle Subject to Some Blockage	deg		11	7.75	7.0	none	7.25	none	6.5	none	6.1
12	Estimated Beam Blockage of OSF-3 by KOUN	dB					+1.5° 8 dB	0	+1.5° 12 dB	0	+1.5° 6 dB	0
13	Estimated Beam Blockage of OSF-3 by NE Water Tower	dB					+4.5° 18 dB	1.5° 18 dB	1.5° 4 dB	0.5° 4 dB	0.5° 4 dB	0.5° 4 dB
14	Estimated Beam Blockage of OSF-3 by SE Water Tower	dB					+1.5° 5 dB	0.5° 5dB	+2.5° 12 dB	+0.5° 18 dB	+2.5° 18 dB	+0.5° 15 dB
15	Estimated Beam Blockage of OSF-3 by MLOS Tower	dB					all, 0.5 dB	all, 0.5 dB	all, 0.4 dB	all, 0.4 dB	all, 0.3 dB	all, 0.3 dB
16	Power Density at Top of OSF-3 Antenna due to KOUN	mW/cm ²					40	3.0	80	4.0	110	3.0
17	Power Density at Bottom of OSF-3 due to KOUN	mW/cm ²					1	0.5	2	0.5	2	0.5
18	Weighted Avg. $\times 3 \times 10^{-5} \times 10^{-3}$	W					3,000	300	6,000	400	8,000	300

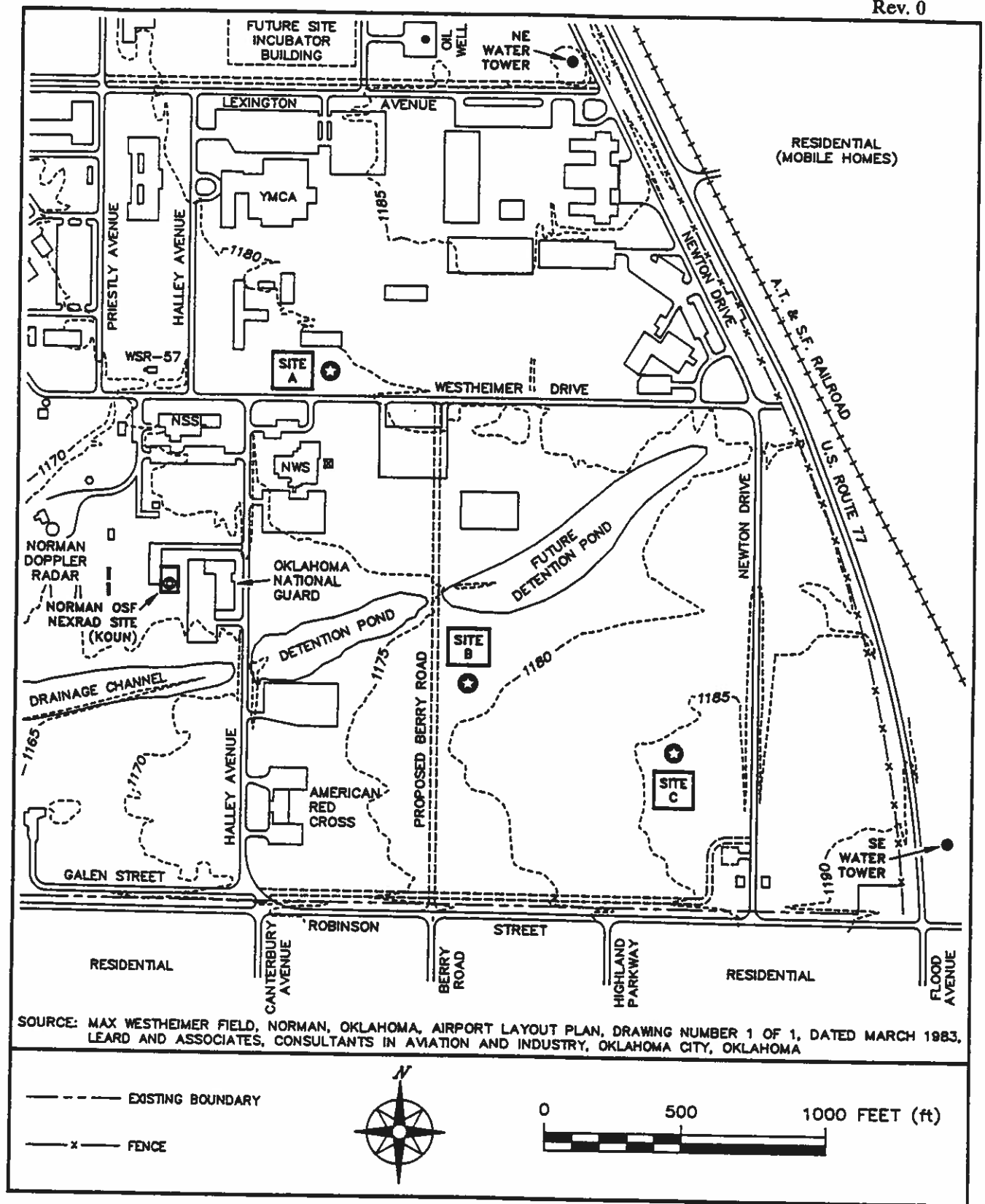


FIGURE 1 POTENTIAL OSF3 SITES AT WESTHEIMER AIRPARK, NORMAN, OKLAHOMA



Operational Support Facility-3
Norman, Oklahoma

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June 8, 1994

~~Sidney Wilkison~~
U.S. Fish & Wildlife Service
Ecological Services
222 S. Houston, Suite A
Tulsa, OK 74127

Dear Mr. Wilkison:

The Next Generation Weather Radar (NEXRAD) Joint System Program Office (JSPO) of the U.S. Departments of Commerce, Defense and Transportation is installing a nationwide network of weather radars to enhance our nation's ability to forecast meteorological events. The NEXRAD network will include over 150 radars serving all 50 states and parts of the Caribbean.

To support operation and maintenance of the NEXRAD system, NEXRAD JSPO plans to install a NEXRAD radar (named OSF3) at the NEXRAD Operational Support Facility (OSF) at Max Westheimer Airpark in Norman Oklahoma. The proposed OSF3 NEXRAD will be the second NEXRAD to be constructed at Max Westheimer, which has had an existing NEXRAD since 1988. Two alternative sites at Max Westheimer Airpark are under consideration for the OSF3 NEXRAD:

Site A is a mowed grassy field in the NE1/4 SE1/4 Section 24, Township 9 North, Range 3 West. This site is at latitude 35°14'16" North and longitude 97°27'34" West (North American Datum (NAD) 1927).

Site B is also a mowed grassy field in the SW1/4 SW1/4 Section 19, Township 9 North, Range 2 West. This site is at latitude 35°14'08" North and longitude 97°27'25" West (NAD 1927).

Both sites are owned by the University of Oklahoma and are available for lease to the NEXRAD JSPO. Figure 1, attached, shows the locations of sites A and B.

The unmanned radar facility will consist of a radar tower and two small equipment shelters within a 60-ft x 100-ft fenced area. The radar will be enclosed within a radome with a diameter of 38 ft mounted on a 30-meter steel tower (see Figure 2, attached). The total height of the radar tower and radome, including the lightning rod, will be 142 ft above ground level. Electronic equipment, a standby electric generator, and an above-ground diesel tank will also be installed within the shelters at the site. Figure 3, attached, is a photograph of a similar NEXRAD facility. The OSF3 NEXRAD will not require water or sewage service. Electric power and telephone service will be provided by connections to existing power and telephone lines at Max Westheimer Airpark.

NO EFFECT FINDING The described action will have no effect on listed species, wetlands, or other important wildlife resources. Date <u>6-20-94</u> Consultation # <u>2-14-94-I-23</u> Approved by <u>Kenneth D. Frazer</u> U.S. FISH and WILDLIFE SERVICE, TULSA, OK

WILDLIFE CONSERVATION COMMISSION
WILLIAM CRAWFORD JOHN S. "JACK" ZINK
CHAIRMAN MEMBER
DON RITTER JOHN D. GROENDYKE
VICE CHAIRMAN MEMBER
JACK FRISBIE HARLAND STONECIPHER
SECRETARY MEMBER
ROBERT L. HUTCHINS MARK PATTON
MEMBER MEMBER



Operational Support Facility-3
Norman, Oklahoma
DAVID WALTERS, GOVERNOR Rev. 0
GREG D. DUFFY, DIRECTOR

DEPARTMENT OF WILDLIFE CONSERVATION

1801 N. Lincoln

P.O. Box 53465

Oklahoma City, OK 73152

PH. 521-3851

July 15, 1994

Mr. James Manidakos
SRI International
333 Ravenswood Ave.
Menlo Park, CA 94025

Dear Mr. Manidakos:

This responds to a letter which we received from your office on June 23, 1994 requesting information on sensitive species and wildlife resource impacts with regard to the following:

Project: Construction of NEXRAD Facilities

Location: SE/4 Section 24, T9N, R2W, Cleveland County
SW/4 Section 19, T9N, R2W, Cleveland County

We have reviewed the information sent to this office regarding the proposed project and our current records of endangered and threatened species. Based on this review, it appears that the project, as described, should not have a significant, long-term impact on state-listed endangered wildlife species or fish and wildlife resources.

For additional information on sensitive species, we recommend that you contact the Oklahoma Natural Heritage Inventory, 2001 Priestly Avenue, Building 605, Norman, Oklahoma 73019. For information on federally listed threatened or endangered species, contact the U.S. Fish and Wildlife Service, Ecological Services, 222 South Houston, Suite A. Tulsa, OK 74127.

We appreciate the opportunity to comment on this project. If we can be of further assistance, please contact our Natural Resources Section at 405/521-4616.

Sincerely,

Mark D. Howery
Natural Resources Biologist

B-23

OKLAHOMA NATURAL HERITAGE INVENTORY
INFORMATION REQUEST RESPONSE

Operational Support Facility-3
Norman, Oklahoma
Rev. 0

Information

Requested By: James Maniakas Jr.

IR LOG: OTH * SBI * 94 * 267

SBI International

QuadCode: 3509724

Date Requested: 7-29-94

County: Cleveland

Center of Location (Township, Range, Section):

T9N, R3W, Sec 24 and T9N, R2W, Sec 19.

REVIEW

1) LISTED OR CANDIDATE ENDANGERED SPECIES (FEDERAL OR STATE SPECIES)

On-site occurrences tagged with v; otherwise, within 1 mile radius.

☒ No known listed or candidate spp. records in database

2) OTHER RARE SPECIES OR COMMUNITIES

On-site occurrences tagged with v; otherwise, within 1 mile radius.

☒ No known other rare spp. or communities records in database

3) PUBLIC LANDS (FEDERAL, STATE OR MUNICIPAL PARKS, REFUGES, MANAGEMENT AREAS)

Occurrence within 1 mile radius.

☒ No known public lands records in database

4) NON-PROFIT MANAGED AREAS (NATURE PRESERVES, REGISTRY NATURAL AREAS)

Occurrence within 1 mile radius.

B-24

☒ No known managed areas records in database

REVIEWERS

Operational Support Facility-3

Norman, Oklahoma

Rev. 0

OKLAHOMA NATURAL HERITAGE INVENTORY

111 E. Chesapeake Street
Norman, OK 73019-0575

TELEPHONE : (405) 325-1985

E-MAIL : ibutler@uoknor.edu

FAX : (405) 325-7702

SIGNATURE(S):

Ian H. Butler

Ian H. Butler, Data Coordinator

DATE OF RESPONSE:

8-4-94

cc: ☐ Ron Suttles, Okla. Dept. of Wildlife Conservation, Oklahoma City
☐ Supervisor, US Fish & Wildlife Service, Tulsa

Attachments (check those attached):

☐ Area Maps
☐ Letter or additional information

THIS REPORT IS BASED SOLELY ON EXISTING INFORMATION IN THE DATABASE. IN THE ABSENCE OF FIELD SURVEYS WE CANNOT STATE WHETHER OR NOT A GIVEN SITE HARBORS RARE SPECIES OR SIGNIFICANT COMMUNITIES. WE SUGGEST YOU ALSO CONTACT THE NATURAL RESOURCES SECTION OF THE OKLAHOMA DEPT. OF WILDLIFE CONSERVATION, AS THEY MAY HAVE SITE SPECIFIC INFORMATION OF WHICH WE ARE UNAWARE.



Oklahoma Historical Society

Founded May 27, 1893

STATE HISTORIC PRESERVATION OFFICE
621 N. ROBINSON, SUITE 375 • OKLAHOMA CITY, OK 73102 • (405) 521-6249

July 9, 1994

Mr. James Manidakos Jr.
SRI International
333 Ravenswood Avenue
Menlo Park, CA 94025

RE: File #1550-94; NEXRAD Project at Westheimer Airpark, Sites A & B

Dear Mr. Manidakos:

We have received and reviewed the documentation submitted concerning the referenced project in Cleveland County.

Examination of historic resource files in this office finds no properties documented within the project area that meet the criteria for listing on the National Register of Historic Places. Our research indicates that there is little likelihood such historic properties will occur.

In addition to review by this office, a review focusing on prehistoric resources by the Oklahoma Archeological Survey is required for determining the presence of National Register quality prehistoric sites. Documentation on any historic archaeological site discovered in the course of archaeological surveys should be submitted to the State Historic Preservation Office for review. This is an integral part of the Section 106 process.

Should the Oklahoma Archeological Survey conclude that there are no prehistoric archaeological sites of National Register quality, and should no historic site have been discovered in the process of survey, the State Historic Preservation Office finds no properties eligible for the National Register of Historic Places within the referenced project boundaries.

Should further correspondence pertaining to this project be necessary, the above underlined file number must be referenced. If you have any questions, please contact Mr. Marshall Gettys, Historical Archaeologist, at 405/521-6249. Thank you.

Sincerely,

Melvena Heisch
Deputy State Historic
Preservation Officer

MH:pm



Oklahoma Archeological Survey

THE UNIVERSITY OF OKLAHOMA

July 26, 1994

Jim Manidakos, Jr.
SRI International
Environmental Assessment Program
333 Ravenswood Ave.
Menlo Park, CA 94025

Re: NEXRAD/Environmental Assessment Program for proposed radar facilities. Legal Description: NE1/4 SE1/4 Section 24 T9N R3W (Site A); SW1/4 SW1/4 Section 19 T9N R2W (Site B); Cleveland County, Oklahoma.

Dear Mr. Manidakos:

The above referenced project has been reviewed by the Community Assistance Program staff of this agency to identify potential areas that may contain prehistoric or historic archaeological materials. The location of your project has been cross-checked with the state site files containing approximately 14,000 archaeological sites which are currently recorded for the state of Oklahoma. No sites are listed as occurring within your project area, and based on the topographic and hydrologic setting, no archaeological materials are likely to be encountered. Thus, an archaeological field inspection is not considered necessary. However, should construction activities expose buried archaeological materials such as chipped stone, tools, pottery, bone, historic crockery, glass, metal items or building materials, this agency should be contacted immediately at (405) 325-7211. A member of our staff will be sent to evaluate the significance of these remains.

This environmental review and evaluation is performed in order to locate, record, and preserve Oklahoma's prehistoric and historic cultural heritage in cooperation with the State Historic Preservation Office, Oklahoma Historical Society. Thank you for your cooperation.

Sincerely,


Marjy Duncan
Staff Archaeologist


Robert L. Brooks
State Archaeologist

:lw

cc: SHPO

B-27

MEMORANDUM

SRI GEOSCIENCE AND ENGINEERING CENTER

TO: Juris Petriceks
FROM: Bill Edson *Bill Edson*
SUBJECT: OSF-3 at Norman, OK - Potential EMIs in MLOS Link

DATE: August 29, 1994
LOCATION: G-234
cc: Patti Burns
Dave Elsner

The existing MLOS tower is almost directly south of the proposed OSF-3 site, and at a distance of about 330 ft. It is self supporting, 151 ft tall, and carries two six-foot parabolic antennas at levels of 86 and 146 ft AGL, both directed toward the Twin Lakes site at an azimuth of $57^{\circ} 32'$. The frequency assigned to the link from Twin Lakes is 1809.5 MHz. The center of the OSF-3 antenna is to be 114 ft AGL, on ground, which is about 5 ft higher than that occupied by the MLOS tower. In 330 ft the center of the lowest (+0.5 deg) beam of OSF-3 will rise about three feet, while that of the +4.5 deg beam will rise about 26 ft. Thus, the elevations of the beam centers will range from 122 to 148 ft AGL at the MLOS tower. The lower parabolic antenna will be well below the lowest OSF-3 beam, but the upper one will be subject to the full intensity of the +4.5 deg beam.

Catalog 38 of the Andrew Corporation lists several 6-foot antennas for this frequency range, Specified midband gain values range from 28.6 to 28.7 dBi, and front-to-back ratios range from 36 to 44 dB. Data on the gain at angles near 60 deg from beam axis are not available, but curves for a 6-ft antenna for the 5.925 - 6.425 GHz band show a response down 37 dB at an angle 15 deg off axis. For the present case a response down 30 dB at 60 deg is taken as a reasonable estimate, which results in a net gain of about -2 dBi for the existing antennas.

At a distance of 330 ft the on-axis peak power density of the OSF-3 beam is about $2,500 \text{ mW/cm}^2$ (from the Titan report). The capture area of an isotropic antenna at the WSR-88D frequency is about 10 cm^2 . Reduced by 2 dB, this is about 6 cm^2 . Thus, the MLOS antenna might capture up to $2,500 \times 6 = 15,000 \text{ mW} = 15 \text{ watts}$. This value is large enough to cause serious EMI and potential for damage to front-end components. Insertion of a suitable lowpass filter in the (coaxial) feedline to the antenna is recommended. Such a filter presents no problem to the reciprocal transmission link, which uses a frequency of 1722.5 MHz.

The lower antenna will be subject to a power level about 30 dB lower, i.e., 15 mW. Power at this level is unlikely to damage front-end components, but might well cause interference. Use of a filter in the feed line of the lower antenna is recommended.

If interference persists in either antenna it can probably be ameliorated by use of a "side blinder" screen of wire mesh. A sheet four-by-eight feet should be large enough.

The existing KOUN is about 700 ft from the MLOS tower, with an antenna about 82 ft AGL, on ground about 5 ft lower than that at the MLOS base. The KOUN beam strikes the backs of the antennas, and for the lowest beam elevation has an intensity of about $1,500 \text{ mW/cm}^2$ at the lowest MLOS antenna. The effective capture area of the lower MLOS antenna is not likely to be more than 20 dB smaller than the sidelobe capture area of 6 cm^2 cited above. On this basis the captured power might be as large as 90 mW. If no interference is being observed it follows that the existing system has a high immunity to EMI.

Polarization

It is known that the WSR-88D uses horizontal polarization, but the polarization used by the MLOS links is not specified. If MLOS uses vertical polarization this difference would provide an additional margin of about 20 dB against EMI.

Beam Blockage

For the OSF-3 system the MLOS tower and its antennas will produce a degree of beam blockage that is roughly equal to that now existing for KOUN. This statement is based on the fact that the separation distances (330 and 700 ft) are both within the near-field range where the beam intensity is nearly independent of distance. Figures 4 and 5 of my memo "Beam Blockage of the WSR-88D Radar", November 4, 1993, indicate that the tower and parabolic reflectors will each create a one-way blockage near 0.5 dB. The distance between the two antennas is large enough so that only one antenna will come into play at a time. Thus, the total predicted one-way blockage is about one dB. Addition of a "side blinder" screen will not appreciably affect this estimate.

Summary and Conclusions

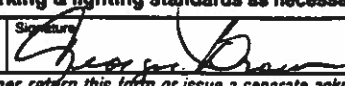
The upper MLOS antenna is likely to receive interfering signals more than 20 dB stronger than those now impinging on either of these antennas. It is quite possible that interference will result. If interference occurs it can be suppressed by additional filtering or by addition of a metal screen to deflect incoming radiation. Beam blockage to OSF-3 will be about the same as that now experienced by KOUN.

*GLI=Grade Level Increase Included
DO NOT REMOVE CARBONS

Please Type or Print on This Form

OSF3, OK RDA NEX

Form Approved OMB NO. 2120-0001

 U.S. Department of Transportation Federal Aviation Administration		Notice of Proposed Construction or Alteration		Aeronautical Study Number _____	
1. Nature of Proposal			2. Complete Description of Structure		
A. Type ☒ New Construction ☐ Alteration *		B. Class ☒ Permanent ☐ Temporary (Duration _____ months)		C. Work Schedule Dates Beginning <u>4th Qtr 95</u> End <u>1st Qtr 96</u>	
* If Alteration, provide previous FAA Aeronautical Study Number, if available: _____					
3A. Name, address, and telephone number of individual, company corporation, etc. proposing the construction or alteration. (Number, Street, City, State, and Zip Code) NEXRAD Joint System Program Office 1325 East West Highway, SSMC-2 Room 15323 Silver Spring, MD 20910 (301) <u>713-1000</u> Area Code Telephone Number			Please describe, on a separate sheet of paper if necessary, the proposed construction or alteration. A. For proposals involving transmitting stations, include effective radiated power (ERP) and assigned frequency of all proposed or modified transmitters on the structure. (If not known, give frequency band and maximum ERP). B. For proposals involving overhead wire, transmission lines, etc., include the size and the configuration of the wires and their supporting structures. C. For all proposals, include site orientation, dimensions, and construction materials of the proposed or altered structure. D. Optional— Describe the type of obstruction marking and lighting system desired for your structure. The FAA will recommend appropriate marking and lighting for the structure in accordance with the standards of Advisory Circular AC 70/7460-1. An FAA marking and lighting recommendation will reflect the minimum acceptable level of conspicuity necessary to warn pilots of the presence of an object. However, the FAA, under certain circumstances, will not object to the use of a system (such as a medium intensity flashing white light system or a dual lighting system) other than the recommended standard. *** See Attachment ***		
3B. Name, address and telephone number of proponent's representative, if different than 3A. above. Georgia Brown, G-223 (Please send questions and response to this address only) SRI International 333 Ravenswood Avenue Menlo Park, CA 94025 (415) <u>859-4842</u> Area Code Telephone Number					
4. Location Of Structure			5. Height and Elevation (to nearest foot)		
A. Coordinates (to hundredths of seconds if known) NAD 83 Latitude <u>35</u> <u>14</u> <u>18</u> . Longitude <u>01</u> <u>27</u> <u>37</u> .		B. Nearest City or Town and State Norman, OK		C. Nearest public or military airport, heliport, flightpark, or seaplane base Westheimer Field (OUN)	
4D. Source of coordinate information for item 4A. above. ☒ USGS 7.5' Quad Chart ☐ Survey ☐ Other Specify _____ Indicate the reference datum of the coordinates, if known. ☐ NAD 27 ☒ NAD 83 ☐ Other Specify _____ Converted Using NADCON		(1). Distance from structure to nearest point of nearest runway approx. 0.4 miles		A. Elevation of site above mean sea level. 1180' MSL + 7' GLI*	
		(2). Direction to 48 South		B. Height of structure including all appurtenances and lighting above ground or water. 142	
		(2). Direction from structure to airport on airport		C. Overall height above mean sea level (A + B) 9' lightning rod inc. 1329	
4E. Describe, on a separate sheet of paper, the location of the site with respect to highways, streets, airports, prominent terrain features, existing structures, etc. Attach a copy of a U.S. Geological Survey quadrangle map 7.5 minute series (or equivalent) showing the construction site. If available, attach a copy of a documented site survey with the surveyor's certification.					
FAILURE TO PROVIDE ALL REQUESTED INFORMATION MAY DELAY PROCESSING OF YOUR NOTICE Notice is required by Part 77 of the Federal Aviation Regulations (14 C.F.R. Part 77) pursuant to Section 1101 of the Federal Aviation Act of 1958, as amended (49 U.S.C. app. § 1501). Persons who knowingly and willfully violate the Notice requirements of Part 77 are subject to a civil penalty of \$1,000 per day until the notice is received, pursuant to Section 901(a) of the Federal Aviation Act of 1958, as amended (49 U.S.C. app. § 1471(a)) as well as the fine (criminal penalty) of not more than \$500 for the first offense and not more than \$2,000 for subsequent offenses, pursuant to Section 902(a) of the Federal Aviation Act of 1958, as amended (49 U.S.C. app. § 1472(a)).					
I HEREBY CERTIFY that all of the above statements made by me are true, complete, and correct to the best of my knowledge. In addition, I agree to obstruction mark and/or light the structure in accordance with established marking & lighting standards as necessary.					
Date 8/15/94		Typed or Printed Name and Title of Person Filing Notice Georgia Brown, Sr. Research Analyst		Signature 	
FOR FAA USE ONLY		FAA will either return this form or issue a separate acknowledgement.			
The Proposal:					
☐ Does not require a notice to FAA. ☐ is not identified as an obstruction under any standard of FAR, Part 77, Subpart C, and would not be a hazard to navigation. ☐ is identified as an obstruction under the standards of FAR, Part 77, Subpart C, but would not be a hazard to navigation. ☐ Should be obstruction ☐ marked ☐ lighted per FAA Advisory Circular 70/7460-1, Chapters _____ ☐ Obstruction marking and lighting are not necessary.					
Supplemental Notice of Construction, FAA Form 7460-2, is required any time the project is abandoned, or ☐ At least 48 hours before the start of construction. ☐ Within five days after the construction reaches its greatest height. This determination expires on _____ unless: (a) extended, revised or terminated by the issuing office; (b) the construction is subject to the licensing authority of the Federal Communications Commission (FCC) and an application for a construction permit is made to the FCC on or before the above expiration date. In such cases the determination expires on the date prescribed by the FCC for completion of construction, or on the date the FCC denies the application. NOTE: Request for extension of the effective period of this determination must be postmarked or delivered to the issuing office at least 15 days prior to the expiration date. If the structure is subject to the licensing authority of the FCC, a copy of this determination will be sent to that agency.					
Remarks _____					
NAD 83 Coordinates (Use these coordinates for any future correspondence with the FAA)		Latitude		Longitude	
Issued in _____		Signature _____		Date _____	

Public Buildings Amendments of 1988 (P.L. 100-678)

Section 6 of the Public Buildings Amendments of 1988 (P.L. 100-678) defines the Federal Government's relationship with local Governments with respect to the construction or alteration of Federally-owned buildings regarding the application of local zoning laws and laws relating to real property development. While the act technically applies only to buildings for which funds will be first appropriated in FY 1990 or after, it does not greatly alter the existing situation stemming from various court decisions, Comptroller General decisions and policies now in effect. Therefore, we plan to follow the procedural requirements now. The following is a paraphrase of Section 6 of the act which adds the following to the Public Buildings Act of 1959 (40 USC 601-616):

SEC 21. COMPLIANCE WITH NATIONALLY RECOGNIZED CODES

- (a) **Building Codes** - Buildings shall be, to the "maximum extent feasible", constructed or altered in compliance with the latest edition of nationally recognized model codes including, but not limited to, building, electrical, fire and life safety, and plumbing codes.
- (b) **Zoning Laws** - Construction or alteration of any building must be done only after consideration of all requirements (other than procedural requirements) of (1) zoning laws and (2) laws relating to landscaping, open space, setbacks, maximum height, historic preservation, esthetic qualities, or other similar local laws which would apply if the construction or alteration was not being done by a Federal agency.
- (c) **Special Rules** - To meet the requirements of the above, the Federal agency must do the following:
 - (1) **State and Local Government Consultation, Review, and Inspections**
 - (A) Consult with local officials in preparation of plans,
 - (B) Upon request, submit plans to local officials for review for a reasonable period of time not to exceed 30 days,
 - (C) Permit inspection by local officials during construction in accordance with the localities customary inspection schedule. The Government can require an inspection schedule and reasonable notice before each inspection.
 - (2) **Limitation on State Responsibilities** - The local Government cannot be obligated to take any action under Paragraph 1 above.
- (d) **State and Local Government Recommendations** - Appropriate local officials may make recommendations concerning measures necessary to meet the requirements of subsections (a) and (b) above or measures which should be taken to take into account local conditions. The Government must give due consideration to the recommendations.
- (e) **Effect of Noncompliance** - No action can be brought and no fine or penalty can be imposed against the Government for failure to meet the requirements of subsections (a), (b), or (c) above or for failure to carry out any recommendation of local officials under subsection (d).
- (f) **Limitation of Liability** - The Government or its contractors cannot be required to pay for any action of the local Government to carry out this section including plan reviews, on-site inspections, issuing building permits, and making recommendations.
- (g) **Applicability** - The act applies to buildings for which funds are first appropriated after FY 1989.
- (h) **National Security Waiver** - The act is not applicable to any building if it's determined that application would adversely affect national security.

KLL 1/11/89

Appendix C

LOCAL POINTS OF CONTACT

Appendix C

LOCAL POINTS OF CONTACT

Owner Representatives

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George Hargett, Administrator (405) 325-7233
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National Weather Service

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Larry Mooney, former Meteorologist-in-Charge
Operational Support Facility
National Weather Service
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Oklahoma Archaeological Survey
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Norman, Oklahoma 73019-0575

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Oklahoma City, Oklahoma 73152

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Max Westheimer Airpark

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(202) 224-4721

Don Nickles (405) 231-4941
1820 Liberty Tower
Oklahoma City, Oklahoma 73102
133 Hart Senate Office Building
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2344 Rayburn House Office Building

Washington, D.C. 20515

(202) 225-6165

Appendix D
ALTERNATIVES CONSIDERED AND REJECTED

Appendix D

ALTERNATIVES CONSIDERED AND REJECTED

The NEXRAD system must meet the mission requirements of four federal agencies: the NWS, the FAA, the AWS, and the NOC. To satisfy those requirements, the preferred site location must optimize the effective radar coverage range or service areas to include population centers, terminal and en route airspace, and military areas, as appropriate. The ideal candidate site must have sufficient elevation and surrounding terrain that allows an unobstructed view of the horizon, particularly in the direction from which severe weather usually approaches the area. Topographical and structural features that decrease the amount of unobstructed radar coverage or increase ground return signals (ground clutter) are avoided. Areas of special environmental or cultural significance, such as wildlife refuges or historic sites, are also included.

The OSF-3 NEXRAD would support the major functions of the NEXRAD OSF: development of software and hardware to maximize the effectiveness of the NEXRAD system, testing of methods and techniques for maintaining and repairing NEXRADs, and troubleshooting problems at individual NEXRAD units throughout the U.S. and the Caribbean. OSF personnel must have convenient physical access to the OSF-3 NEXRAD to perform these functions in an efficient manner. In particular, OSF staff frequently respond to telephone calls from technicians maintaining NEXRADs and suggest methods to repair remote NEXRADs. It will often be necessary for the OSF staff to use the OSF-3 NEXRAD to explore diagnostic and repair techniques to respond to those calls. Therefore, a site near the OSF administrative building at Max Westheimer Airpark was sought for the OSF-3 NEXRAD.

A total of nine possible sites at Max Westheimer Airpark were considered for OSF-3 NEXRAD. Because the OSF-3 NEXRAD could cause EMI with the existing KOUN NEXRAD at the airfield, possible sites were carefully investigated. To reduce the potential for EMI between the KOUN and OSF-3 NEXRADs, the two should be at different elevations. Terrain at Max Westheimer Airpark is essentially flat; thus, the OSF-3 NEXRAD should be on an at least 10-m tower or higher tower than the KOUN NEXRAD which has a 20-m tower. Because a 5-m or 10-m tower would have the potential to cause television interference, have the potential for perceived RF radiation hazard, and would result in blockage of the OSF-3 NEXRAD signal by nearby buildings, a 30-m tower (the tallest available) was selected. However, the 30-m OSF-3 NEXRAD tower would result in some blockage of the KOUN NEXRAD signal. To minimize the coverage loss to the KOUN NEXRAD caused by that blockage, it would be desirable to align the OSF-3 NEXRAD with existing water towers in the area which already block the KOUN NEXRAD signal. Based on these considerations, two of the nine possible sites identified by OSF were found to be most advantageous. The preferred site is

aligned with the north water tower at Max Westheimer Airpark and is analyzed in this report. The alternative site, aligned with the south water tower, and the No-Project Alternative are discussed below.

Alternative Site

This site is located at Max Westheimer Airpark in the southwest quadrant of Section 19, T. 9 N., R. 2 W., and is about 1,250 ft southeast of the preferred site. The site is a grassy field adjacent to the proposed extension of Berry Road and about 750 ft north of Robinson Street. This site is about 1,000 ft from the OSF building, as compared to 300 ft for the preferred site. Environmental conditions are similar as at the preferred site. However, land in the vicinity of this site is planned for commercial development and NEXRAD would be visually less compatible with future uses at this site than at the preferred site (George Hargett, personal communication, May 1994). In addition, existing utilities lines are further from this site than for the preferred site, and utility extensions would be longer and more costly. For these reasons, the alternative site was not chosen for the OSF-3 NEXRAD.

No-Project Alternative

Under this alternative, OSF-3 NEXRAD would not be installed at Max Westheimer Airpark, and the ability of the OSF to support the nationwide NEXRAD system would be diminished. The minor commitment of resources and land disturbance required to install OSF-3 NEXRAD would not occur. However, environmental effects would be minimal and would be outweighed by the benefits to be realized by implementation of the proposed action. Therefore, this alternative was rejected.